

Human-Machine Partnerships in Surgical Robot Systems

Russell H. Taylor

John C. Malone Professor

Department of Computer Science, with joint appointments

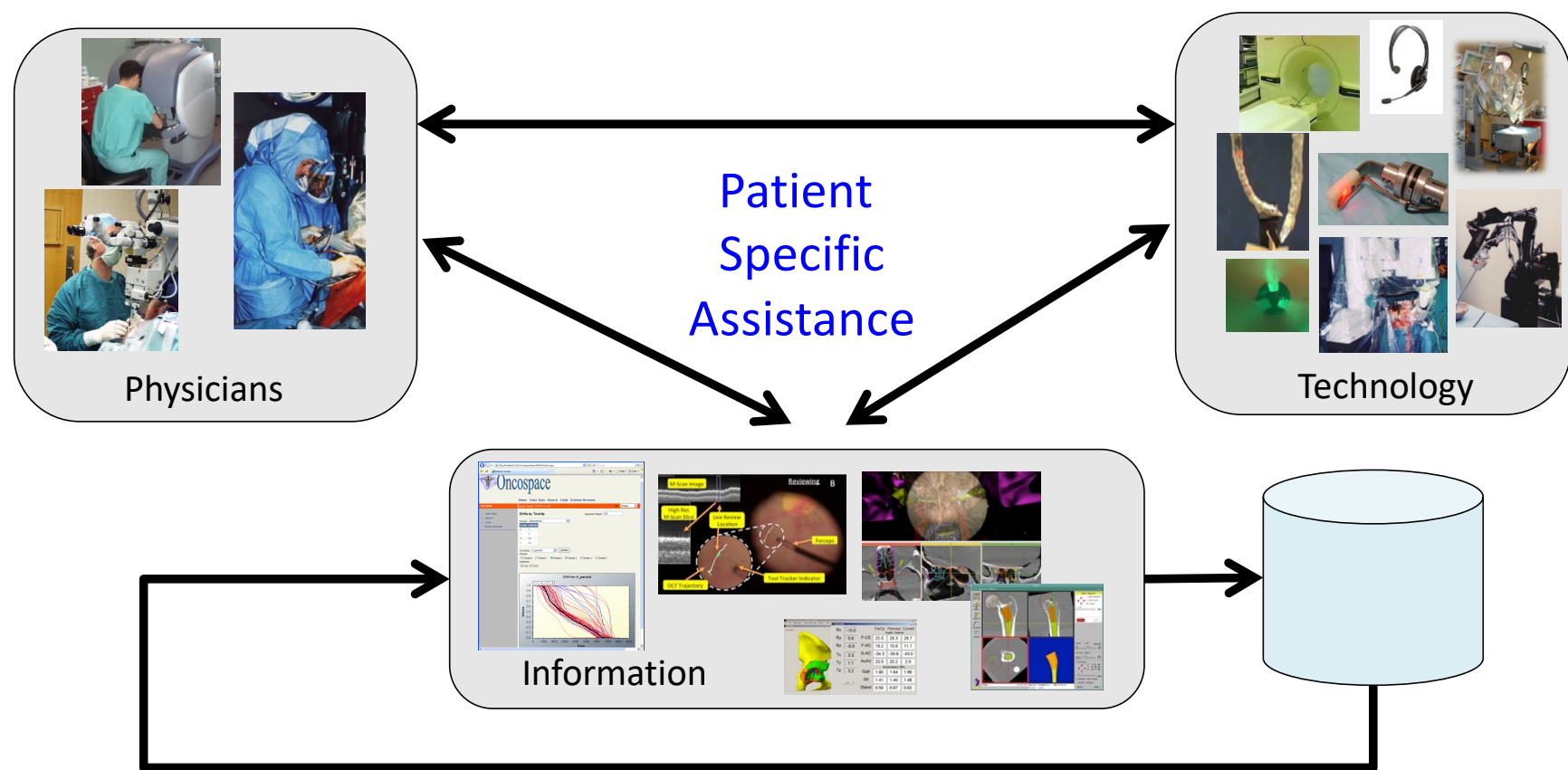
In Mechanical Engineering, Radiology, and Surgery

Director, Laboratory for Computational Sensing and Robotics

The Johns Hopkins University

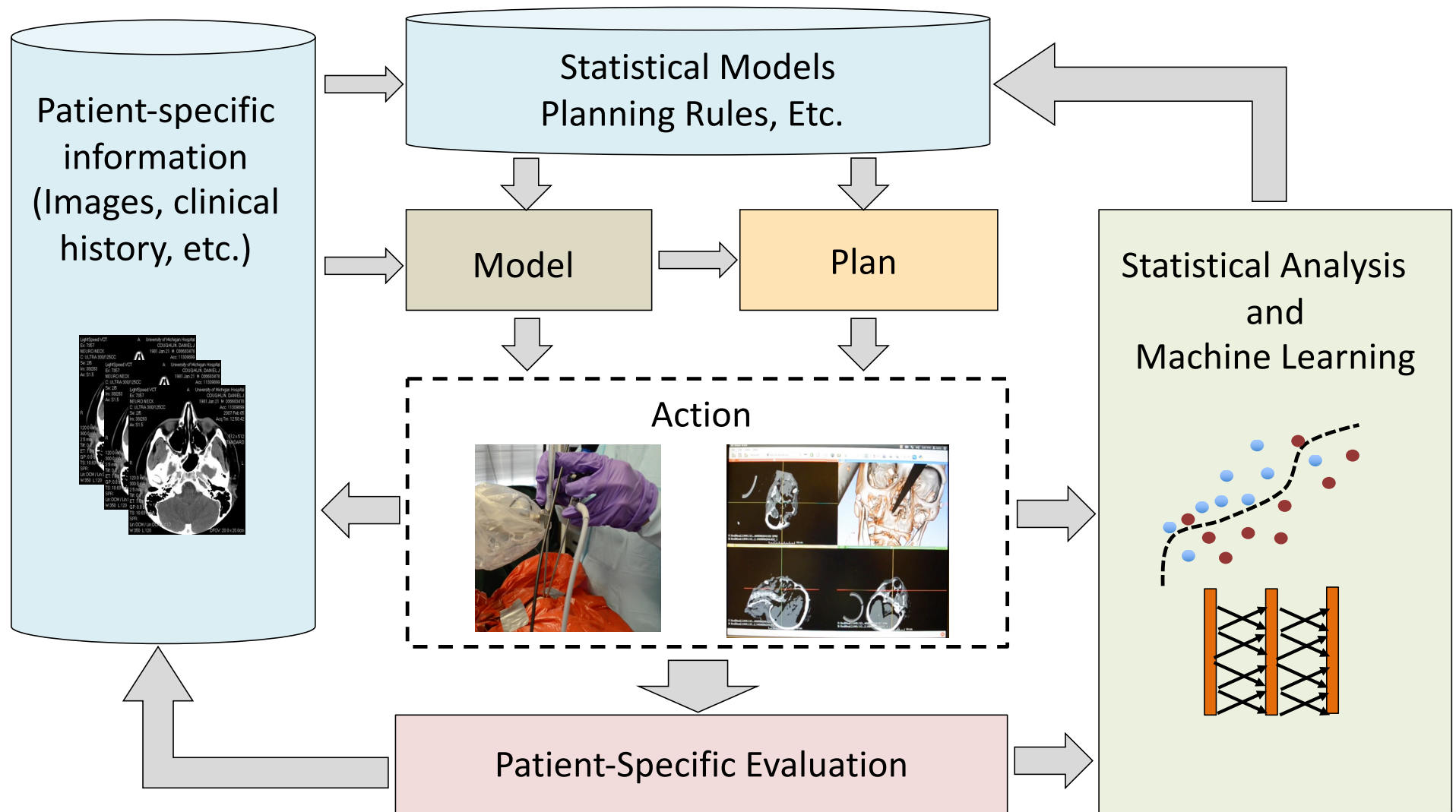
rht@jhu.edu

Human-machine partnership to fundamentally improve interventional medicine



Statistical Process Improvement



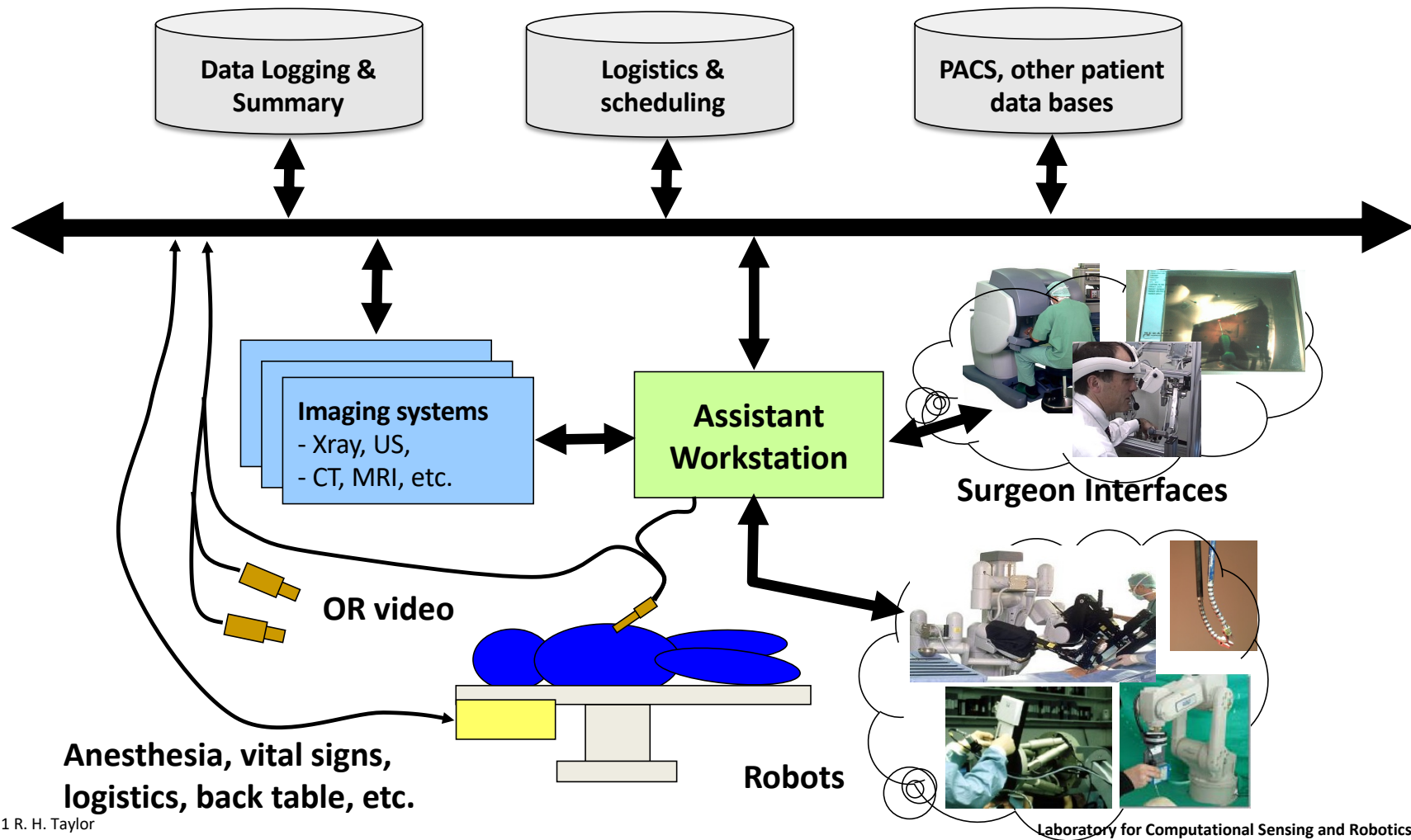


Complementary capabilities

Humans	Machines
• At human scale: – Excellent dexterity – Reasonable precision – Good visual acuity – Excellent haptic senses – Vision and haptic manipulation • Limited manipulation inside the body • Susceptible to ionizing radiation • Excellent situational & task awareness – Challenges for non-visual & haptic senses • Excellent memory and learning ability • Excellent judgment & ability to improvise	• At multiple scales: – Reasonable dexterity – Capable of very precise motions – Geometric accuracy – Multiple sensors; exquisite sensitivity – Sensor-based control • Able to operate inside the body • Can be hardened to x-rays • Limited situational & task awareness – Improving ability to fuse multiple sensors • Increasing ability to store and generalize large amounts of data • Limited judgment and improvisation skills



Information-Intensive Interventional Suite



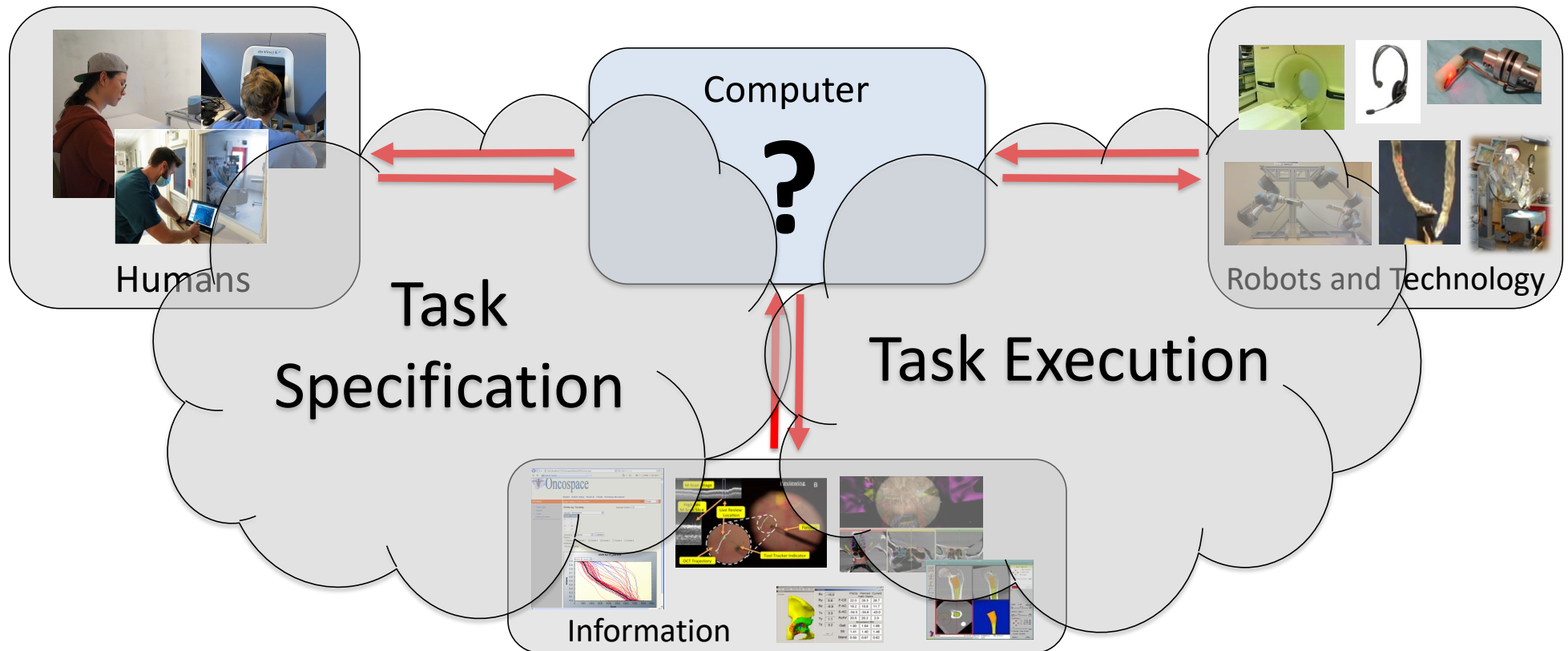
Surgical Robot “Autonomy”

- Human-robot partnerships in which computer makes some or all of the decisions.
- Many taxonomies expressing various “levels” of autonomy
- But there are really two key concerns:
 - Unambiguously specifying what the robot is to do in ways that the computer can understand
 - Ensuring that the computer can cause the the robot to do what it is supposed to do and not something else

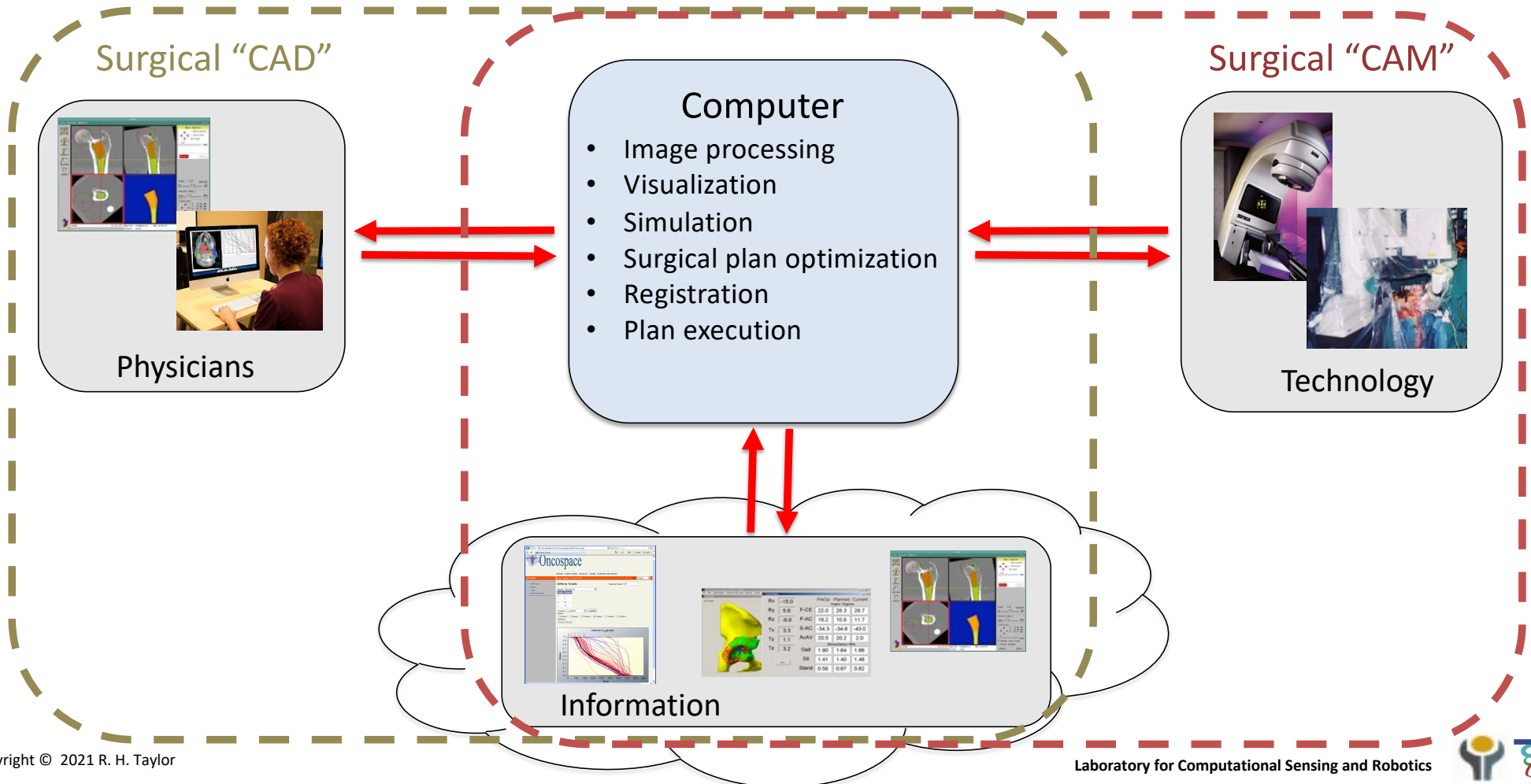


G.-Z. Yang, J. Cambias, K. Cleary, E. Daimler, J. Drake, P. E. Dupont, N. Hata, P. Kazanzides, S. Martel, R. V. Patel, V. J. Santos, and R. H. Taylor, "Medical robotics—Regulatory, ethical, and legal considerations for increasing levels of autonomy [Editorial]", *Science Robotics*, vol. 2- 4, p. eaam8638, 15 March, 2017. 10.1126/scirobotics.aam8638

One key issue: mediating between human intention and action

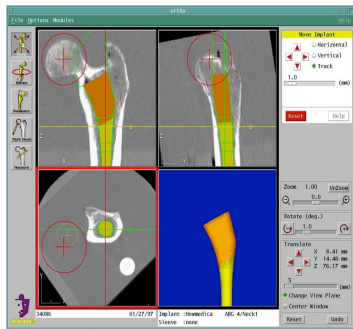


Surgical “CAD/CAM”



Early Examples of Surgical “CAD/CAM” with Robots

Robodoc Total Joint Replacement



1992

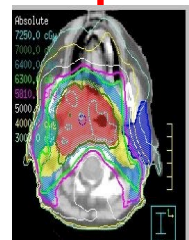
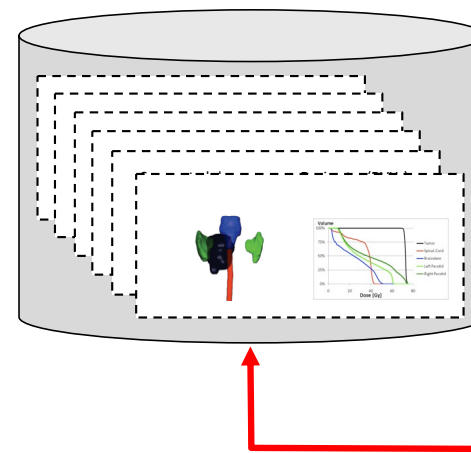
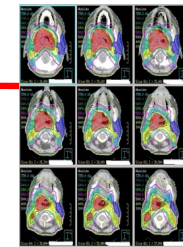
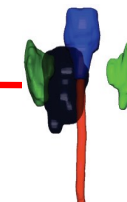


Taylor, Paul, Kazanzides, *et al.*

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Radiation Therapy Systems

2009

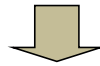
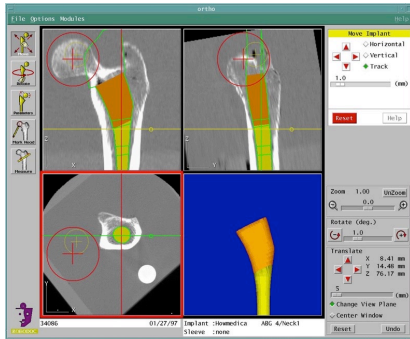


Taylor, McNutt, Kazhdan, Patrucio, Wu, *et al.*

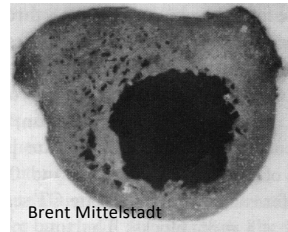
Laboratory for Computational Sensing and Robotics



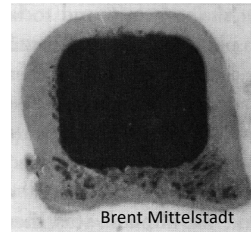
Robotic Joint Replacement Surgery



Taylor, Kazanzides, Paul, Mittelstadt, *et al.*



Manual Surgery



Robotic Surgery

Planning:

- Initially interactive graphics with CT images
- Subsequently
 - Automate segmentation
 - Statistics based planning

Execution

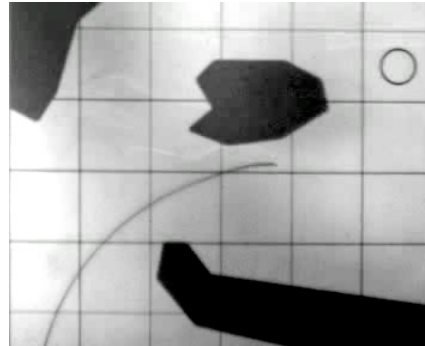
- Combination of hand guiding and autonomous machining bones
- Initially mechanical location of fiducials for registration
- Subsequently
 - ICP-based registration
 - Image-based registration



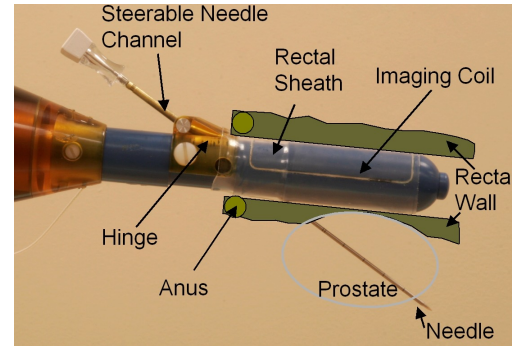
Image-guided needle placement



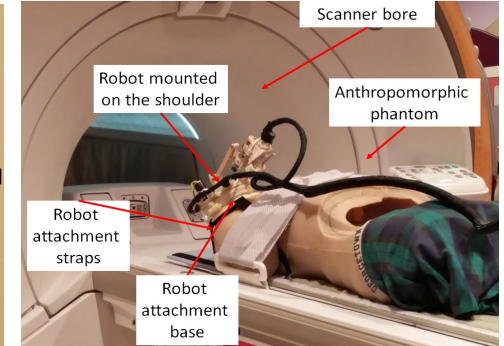
Masamune, Fichtinger, Iordachita, ...



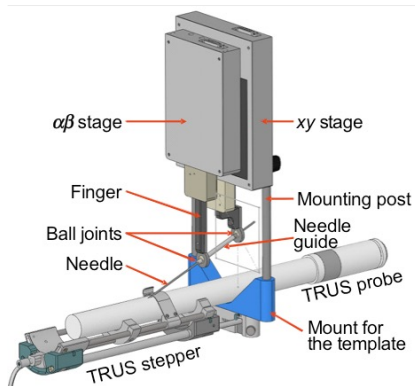
Okamura, Webster, ...



Krieger, Fichtinger, Whitcomb, ...



Monfaredi, Sharma, Kim, Iordachita, Cleary



Fichtinger, Kazanzides, Burdette, Song ...



Taylor, Masamune, Susil, Patriciu, Stoianovici, ...



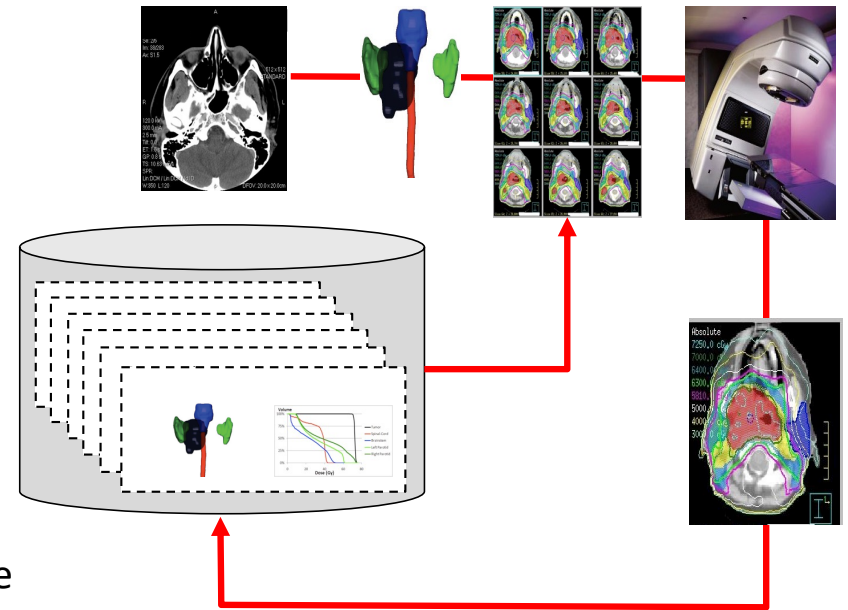
Iordachita, Fischer, Hata...



Iordachita, Fischer, Hata...

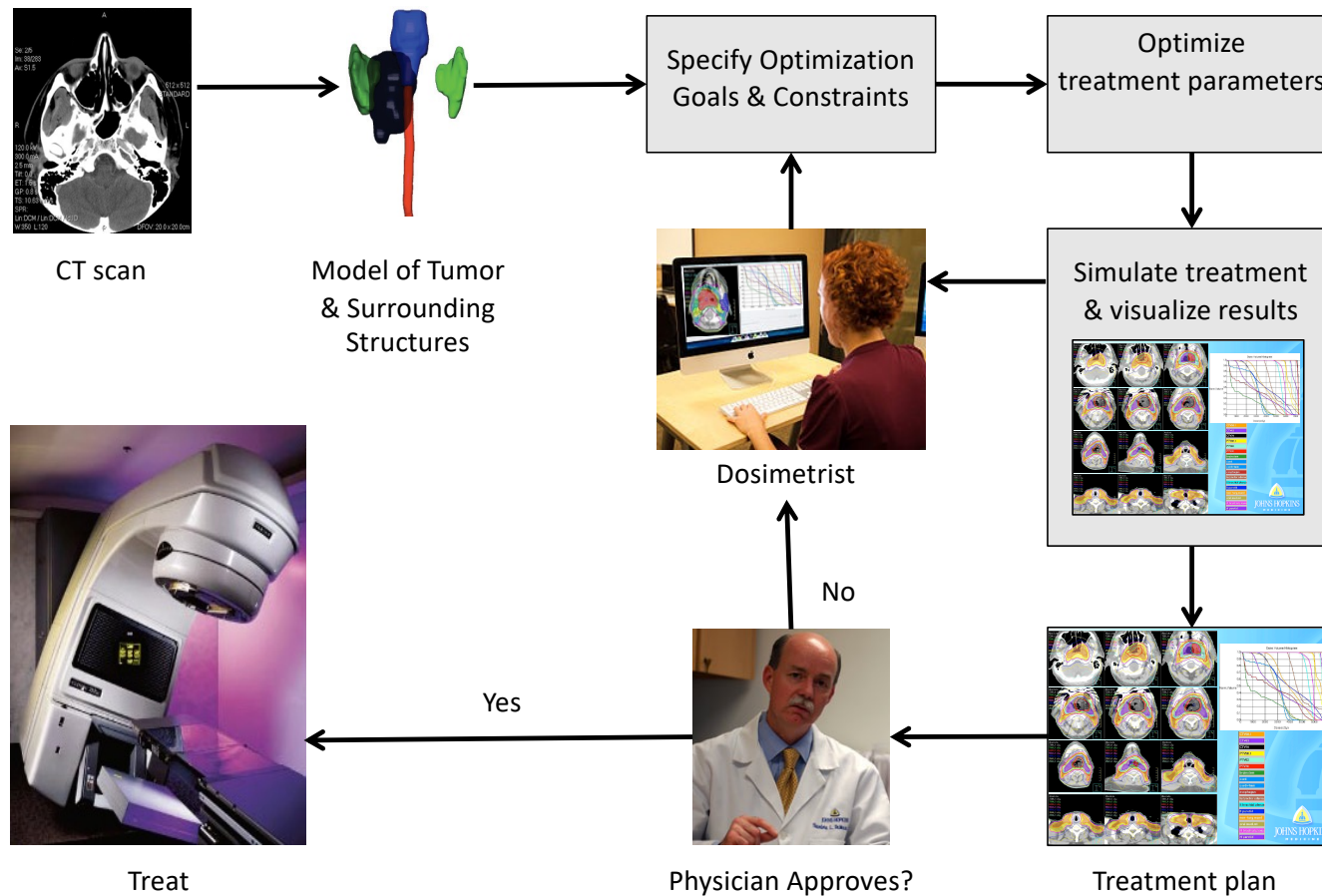
Example: External Beam Radiation Therapy Systems

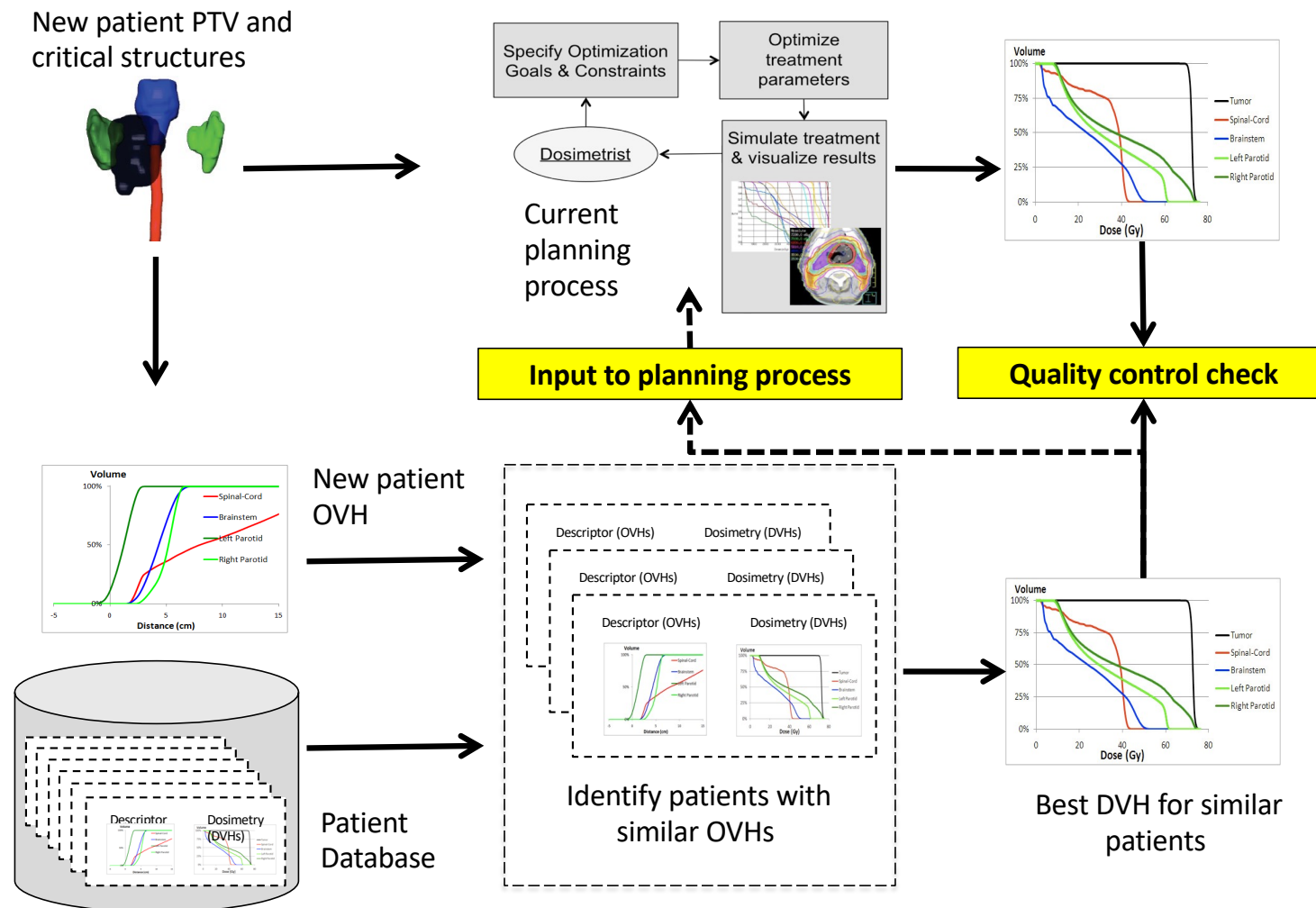
- “Robotic” systems since at least 1980s
- Task Specification
 - Planning of radiation pattern from CT
 - Typically human-machine process involving optimization + simulation
- Task Execution
 - Very careful and accurate machine calibration & verification
 - Registration to patient
 - Machine delivers beams of radiation from multiple angles
- Challenges/Opportunities
 - Adaptation to patient changes/motion
 - Experience-based planning to optimize outcomes
 - The “usual” (system integrity, etc.)



JHU Faculty: Todd McNutt, Russell Taylor, Mischa Kazhdan, Ilya Shpitser, Sauleh Siddiqui

Conventional Radiation Therapy Planning





T. McNutt, B. Wu, M. Kazhdan, P. Simari, A. King, R. Jacques, J. Wong, R. Taylor



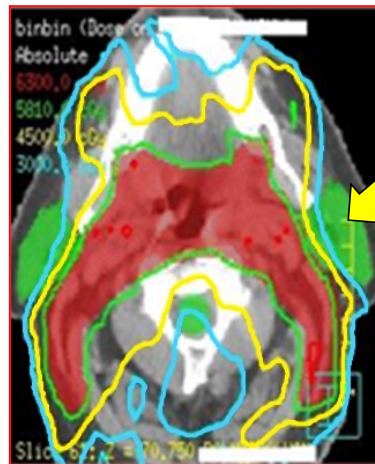
Sample automated radiation planning result

T. McNutt, A. Patriciu, B. Wu, R.. Taylor *et al.*

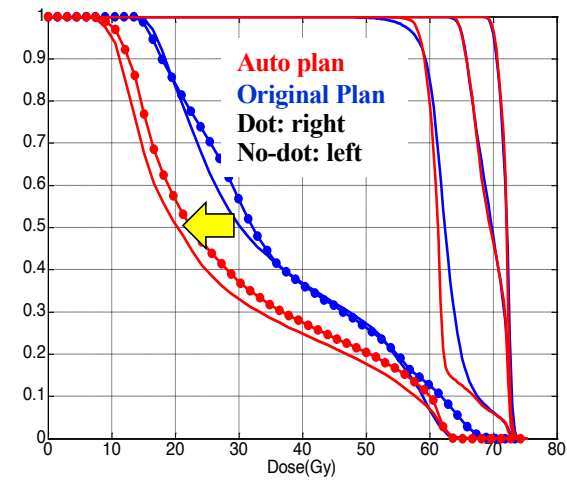
Original plan



Automated plan



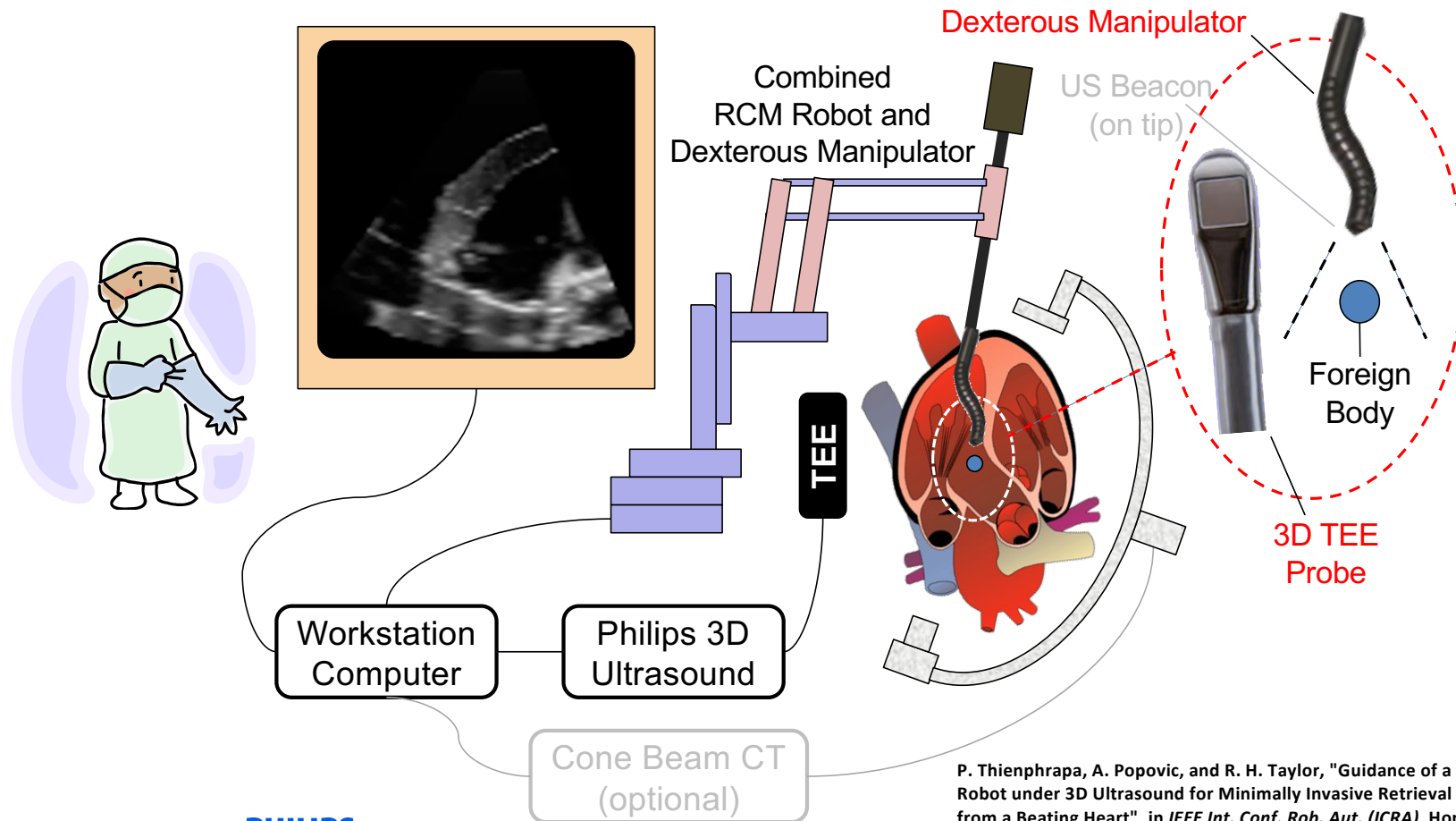
30% reduction in dose to parotids



	brain (Gy) (max)	Brainstem (Gy) (max)	Cord4mm (Gy) (max)	L inner ear (Gy)(mean)
original	61.25	54.58	41.75	57.18
re-plan	56.33	46.48	37.89	43.72
	R inner ear (Gy) (mean)	mandible (Gy) (max)	larynx for edema (V50)	esophagus (Gy)(max)
original	40.57	66.58	61%	63.74
re-plan	38.38	63.78	59%	61

Beating Heart MIS with 3D US Guidance

Paul Thienphrapa, Aleksandra Popovic, Russell Taylor



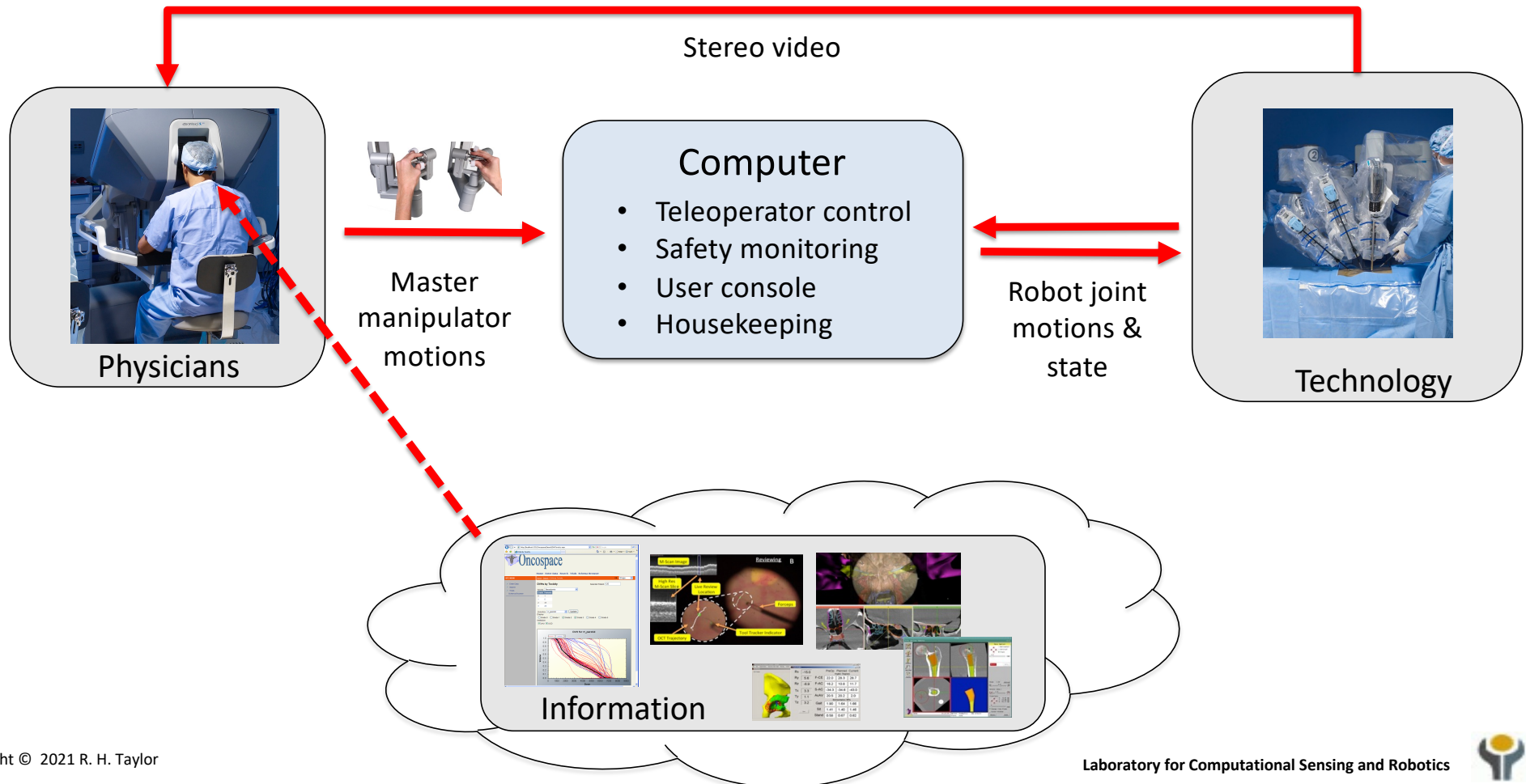
P. Thienphrapa, A. Popovic, and R. H. Taylor, "Guidance of a High Dexterity Robot under 3D Ultrasound for Minimally Invasive Retrieval of Foreign Bodies from a Beating Heart", in *IEEE Int. Conf. Rob. Aut. (ICRA)*, Hong Kong, May 31-Jun 6, 2014. pp. 4869-4874.

Laboratory for Computational Sensing and Robotics

PHILIPS



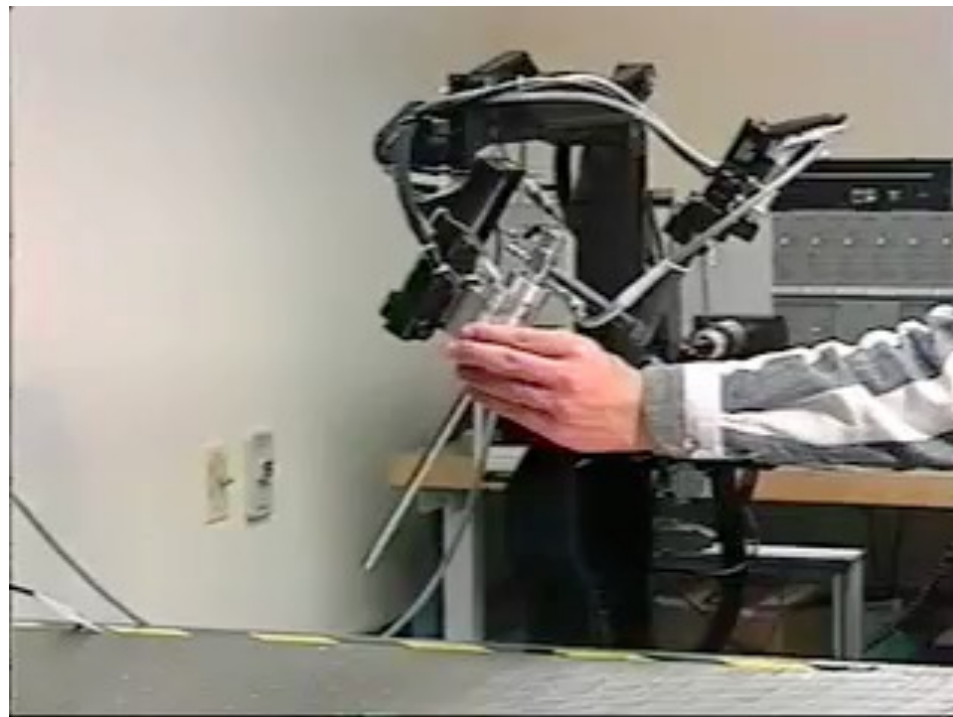
Current dominant paradigm for interactive surgery



Steady Hand Manipulation



Robodoc Canine Surgery, 1990



LARS Robot in IBM Lab, *ca.* 1993

An Integrated High-dexterity Cooperative Robotic Assistant for Intraocular Micromanipulation

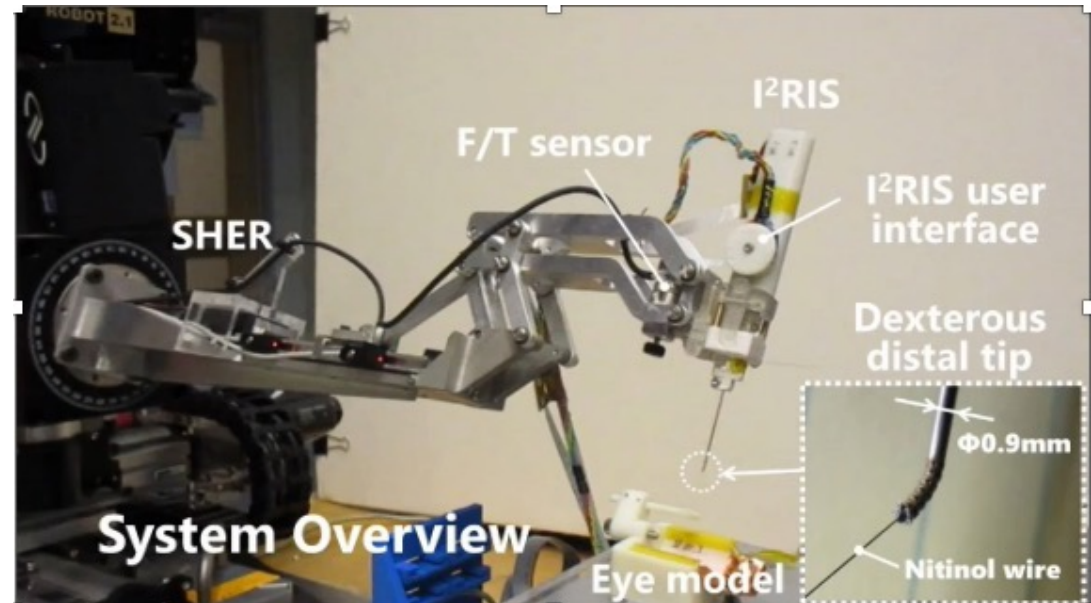
Makoto Jinno

School of Science and Engineering, Kokushikan University, Japan

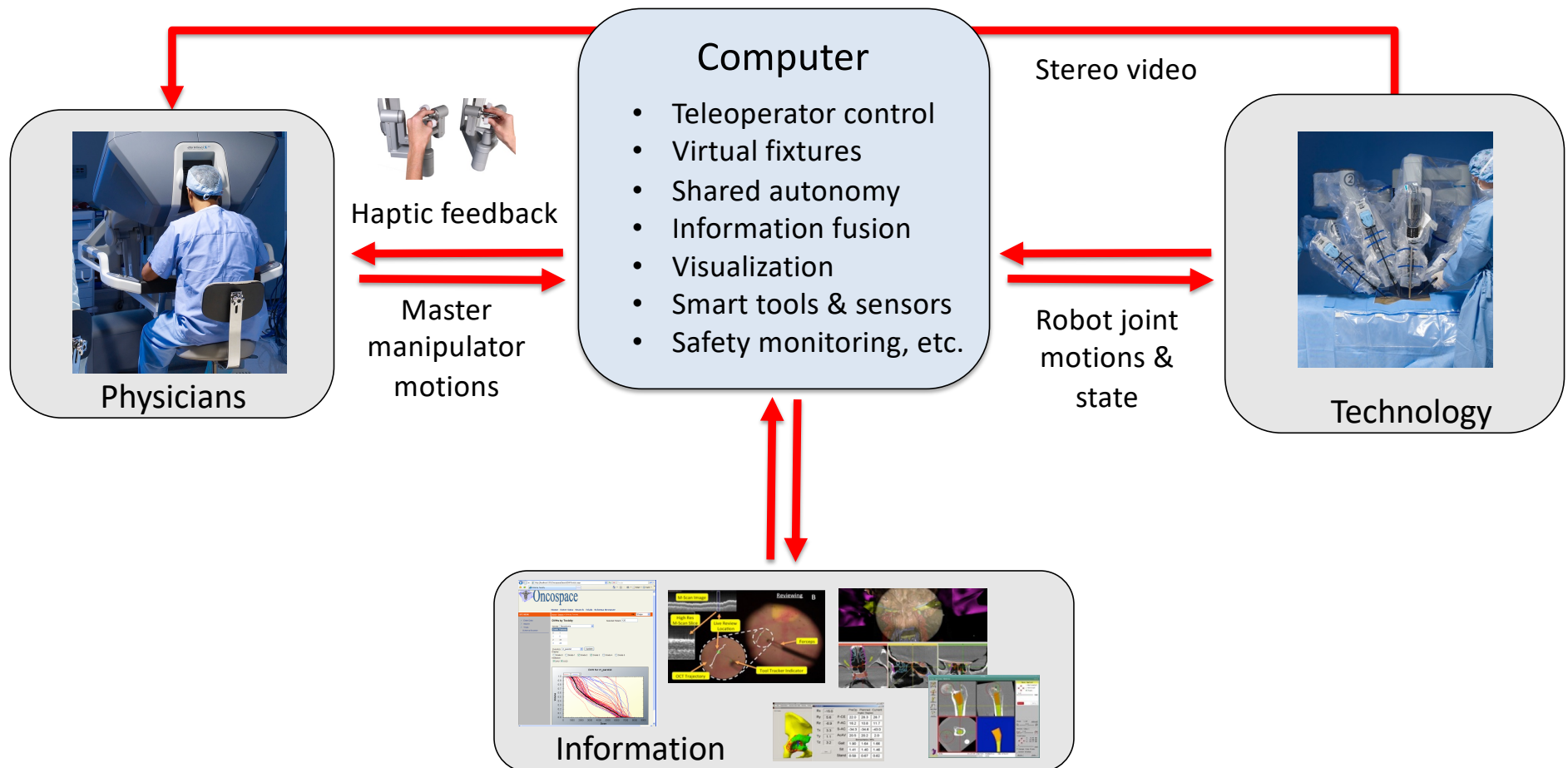
Gang Li, Niravkumar Patel and Iulian Iordachita

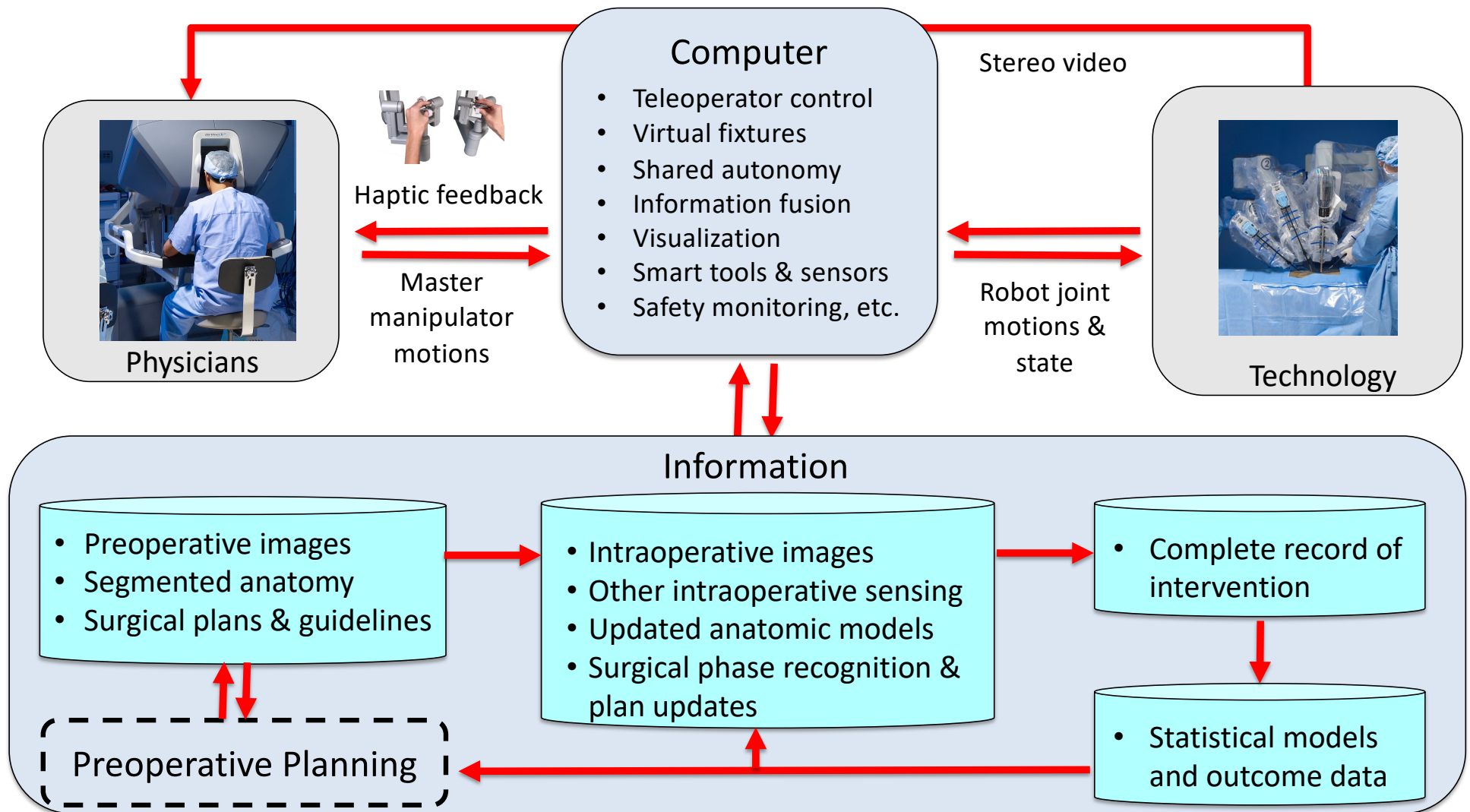
WSE, LCSR, Johns Hopkins University, USA

- This system comprises an improved integrated robotic intraocular snake (I²RIS) and the Steady-Hand Eye Robot (SHER).
- Two user interfaces (joystick and tactile switch) for the I²RIS were developed and evaluated.
- The usability experiments using eye models indicated high-dexterity for either user interface.



Emerging paradigm (shared autonomy & assistant modes)

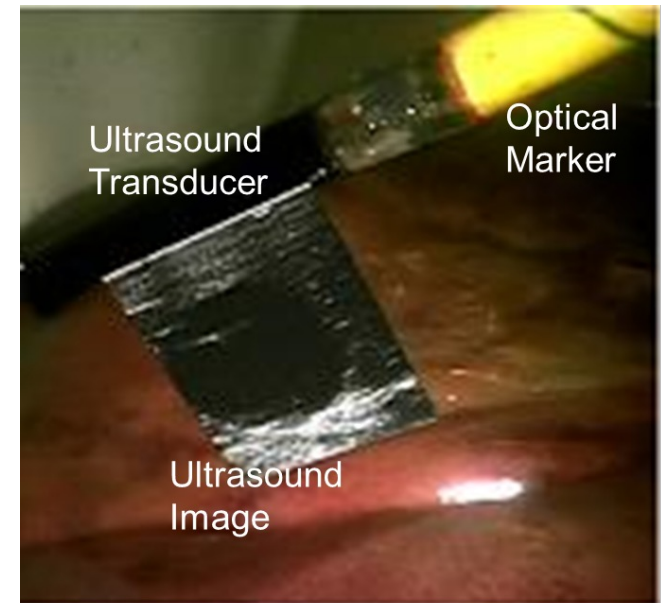




ULTRASOUND PALPATION



Seth Billings, Nishikant Deshmuk, Hyun-Jae Kang, Russ Taylor, Emad Boctor (2011)



Leven J. et al., "DaVinci Canvas: A Telerobotic Surgical System with Integrated, Robot-Assisted, Laparoscopic Ultrasound Capability". MICCAI 2005, 811-818

Emerging paradigm (shared autonomy & assistant modes)

Physicians



Technology

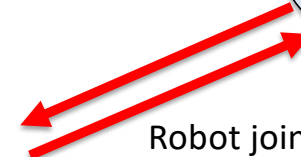
Computer

- Cooperative control
- Virtual fixtures
- Shared autonomy
- Information fusion
- Visualization
- Smart tools & sensors
- Safety monitoring, etc.

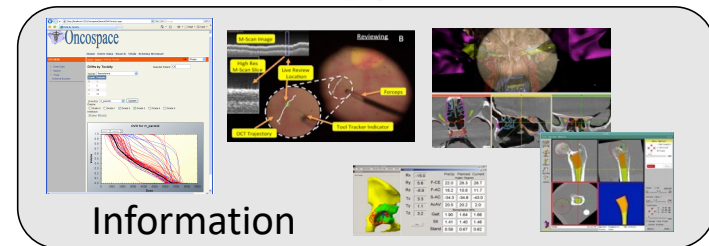
Haptic feedback



Surgeon
Hand Forces



Robot joint
motions & state

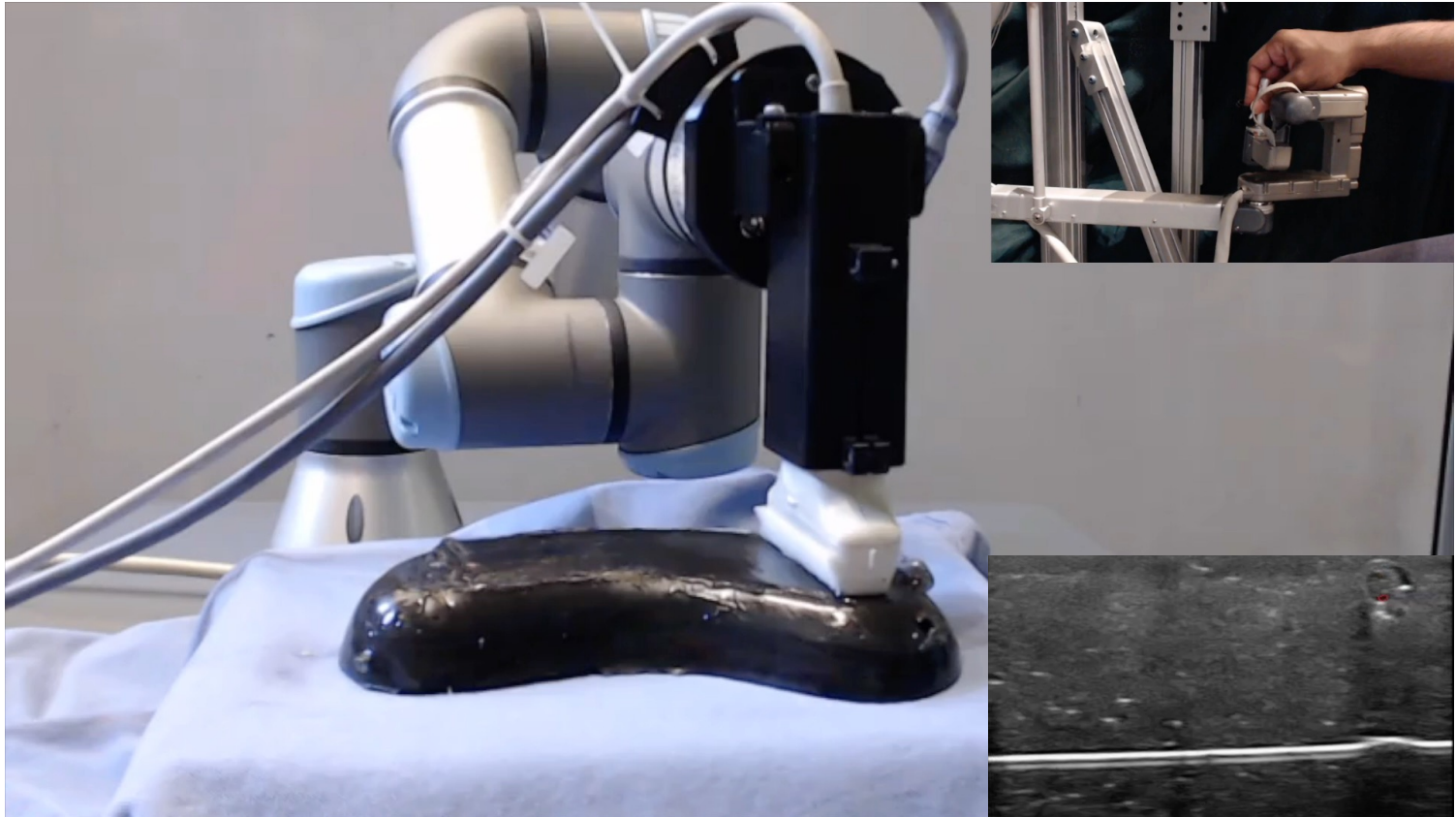


Information

Laboratory for Computational Sensing and Robotics



ULTRASOUND PALPATION



Co-robotic Ultrasound Imaging

Emad Boctor, R. Taylor, *et al.*



Wrist flexion and “pinch” grip



Trunk and neck twist



Trunk flexion

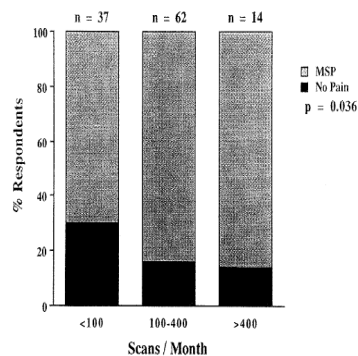


Figure 3 Bar graphs comparing the prevalence of musculoskeletal pains (MSP) with the average number of echo studies performed each month. The number of scans are subdivided due to the wide range of responses. Ninety percent of sonographers performing over 100 scans/month had MSP ($p = 0.036$ for both categories).

Challenges on ultrasound imaging:

- Limited image quality and field of view
- Limited reproducibility
- High user dependency
- Work related musculoskeletal pain (MSP) affect 63 - 91% of sonographers

Robotic Ultrasound Solution

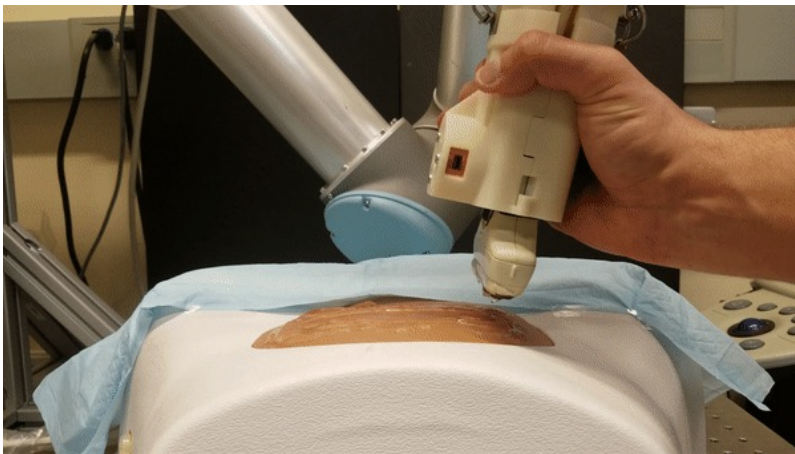
Smith A. C., et al., J Am Soc Echocardiog, 1997. Coffin C., Reports in Medical Imaging, 2014.



Image-based automated tracking of lesion to stabilize view in biopsy

- **Robotic arm:**
 - force control, accuracy, repeatability
- **Co-robotic ultrasound**

In a biopsy procedure, **physiological motions** of the target is another problem to solve.



“Hand-over-hand control”

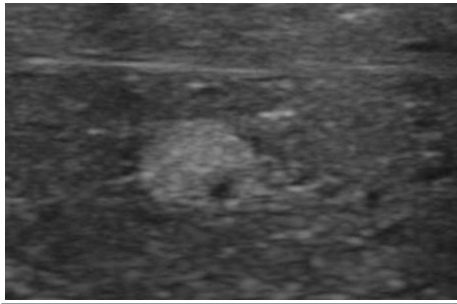


Respiratory motion

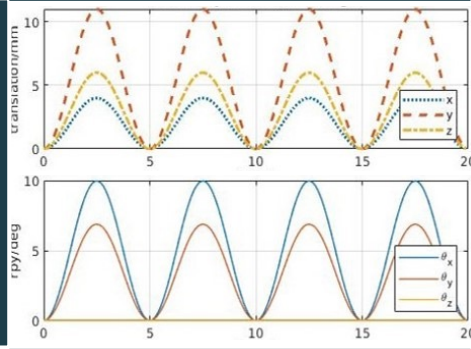
T. Xie, M. Shahbazi, Y. Wu, R. H. Taylor, and E. M. Boctor, "Stabilized ultrasound imaging of a moving object using 2D B-mode images and convolutional neural network", in *Proc.SPIE*, 2020 <https://doi.org/10.1117/12.2550198> 10.1117/12.2550198.

Image-based automated tracking of lesion to stabilize view in biopsy

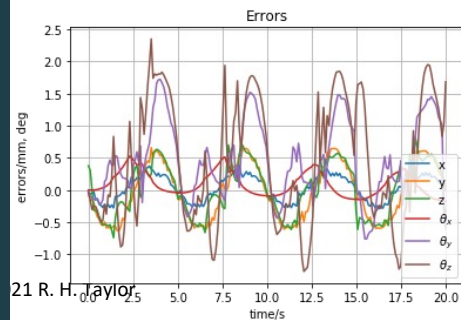
Target view



Target motion

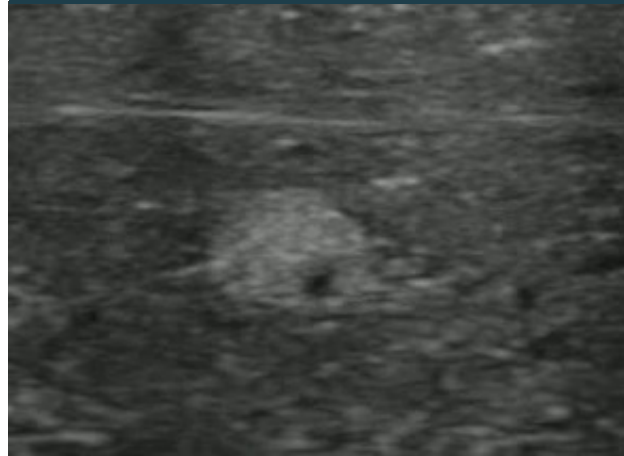


Errors over time

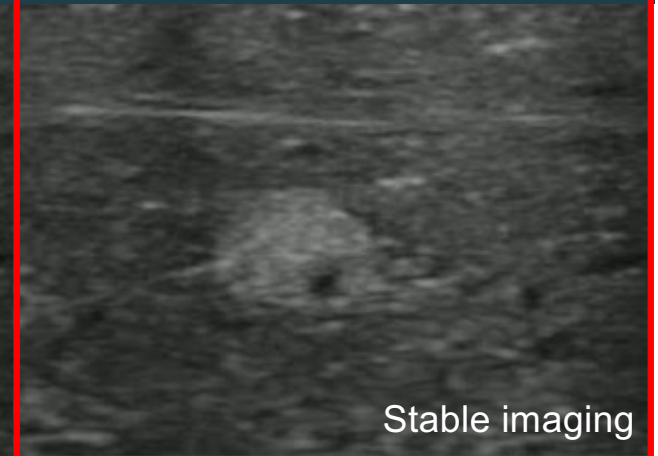


- **Target motion:**
 - 4, 11 and 6 mm along axial, lateral and elevational axes
 - 10 and 6 degree about x and y axes (w/o in-plane rotation)
- **Error:**
 - Translations: less than 0.7 mm
 - Rotations: less than 2 degree
- No fine tuning. Same linear probe used for CNN training.

No tracking



With tracking



Stable imaging

T. Xie, M. Shahbazi, Y. Wu, R. H. Taylor, and E. M. Boctor, "Stabilized ultrasound imaging of a moving object using 2D B-mode images and convolutional neural network", in *Proc.SPIE*, 2020 <https://doi.org/10.1117/12.2550198> 10.1117/12.2550198.

More Details



Stabilized Ultrasound Imaging of a Moving Object Using 2D B-mode Images and a Convolutional Neural Network

Tian Xie^a, Mahya Shahbazi^a, Yixuan Wu^b, Russell H. Taylor^{a, b}, Emad Boctor^{a, b, c}

^a Laboratory for Computational Sensing and Robotics, Johns Hopkins University
^b Department of Computer Science, Johns Hopkins University
^c Russell H. Morgan Department of Radiology and Radiological Science, Johns Hopkins Medical Institutes



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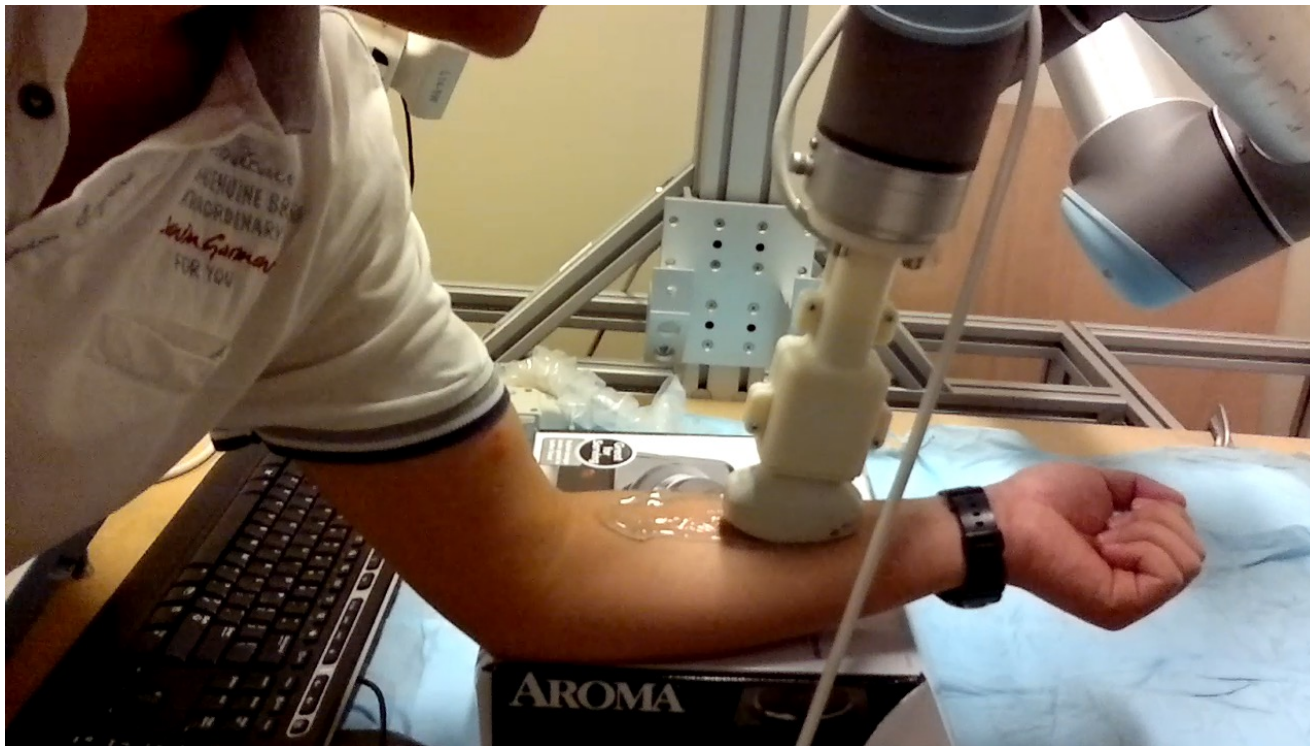


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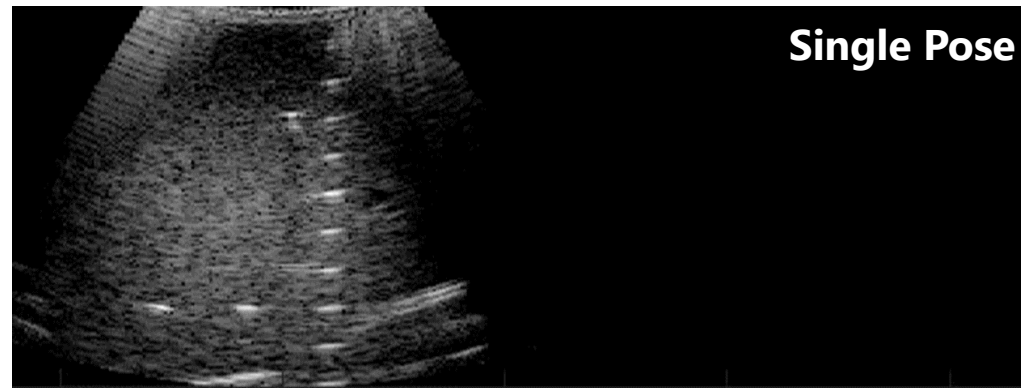
Synthetic-Tracked Aperture Ultrasound (STrAtUS) Imaging Using Robotic Guidance

Emad Boctor, *et al.*

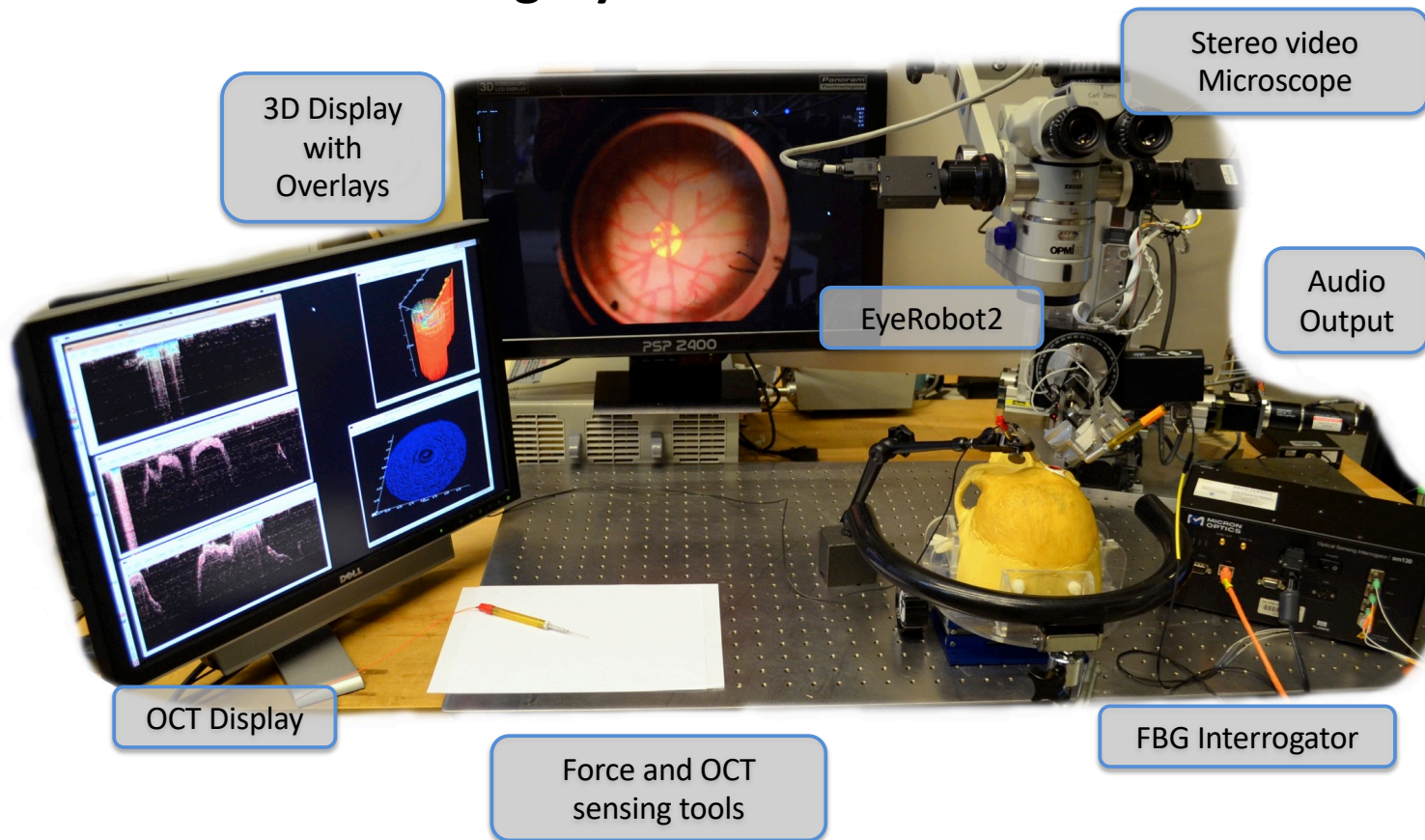


Synthetic-Tracked Aperture Ultrasound (STrAtUS) Imaging Using Robotic Guidance & Virtual Fixtures

Emad Boctor, *et al.*

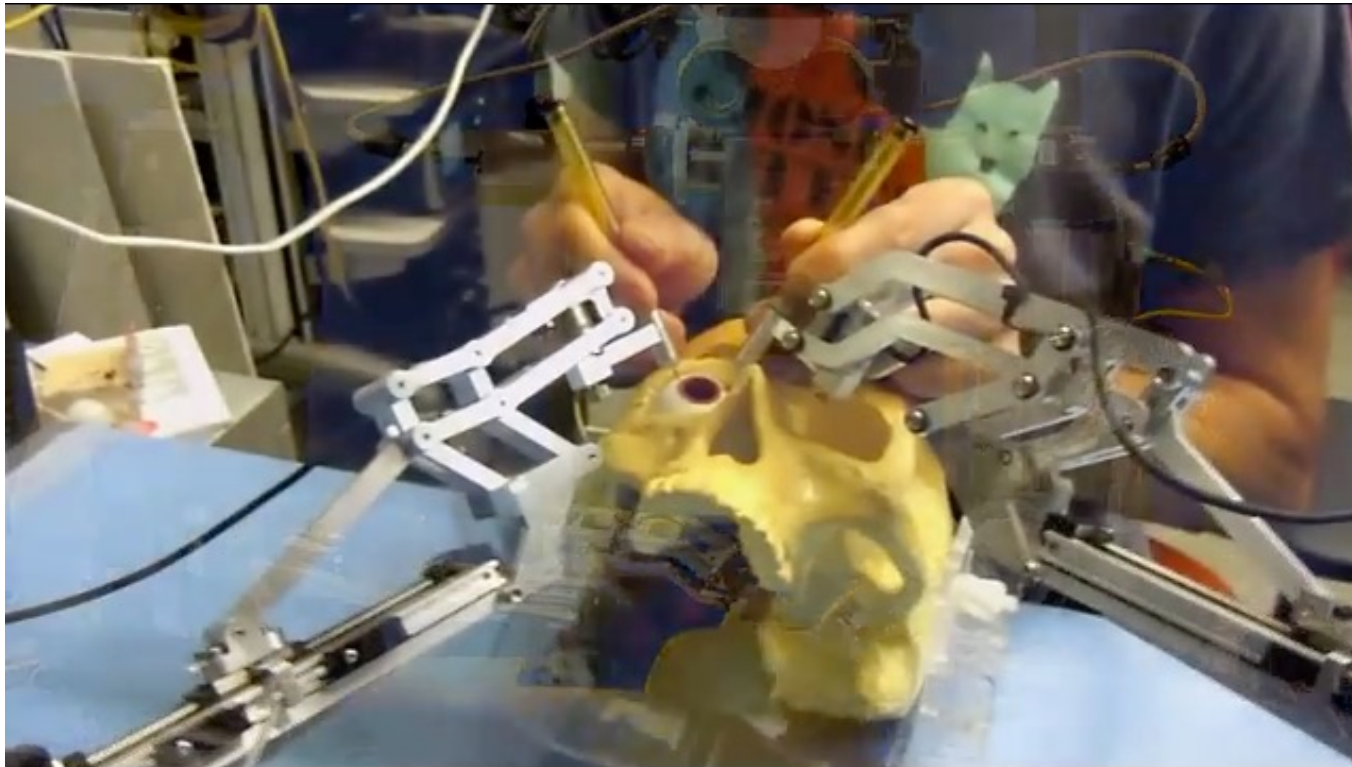


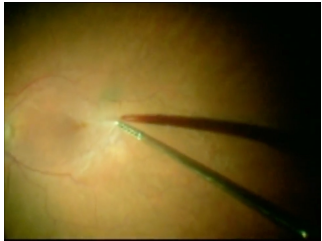
Microsurgery Assistant Workstation



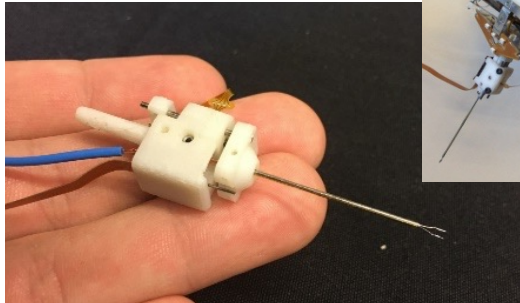
Bilateral Robot-Assisted Retinal Surgery

Iulian Iordachita, *et al.*

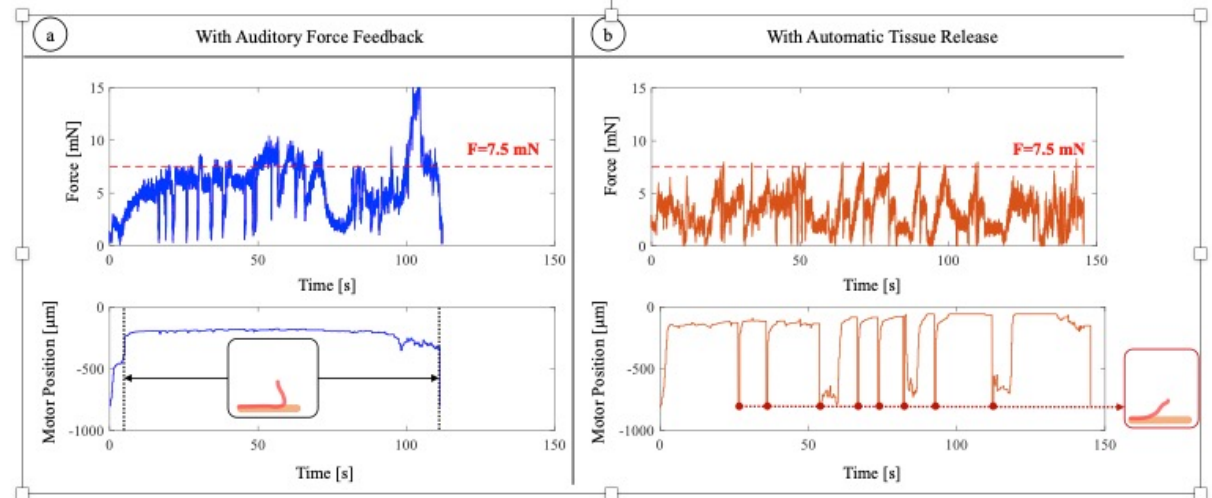
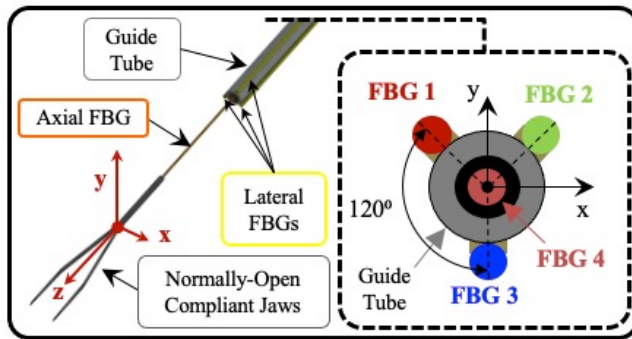
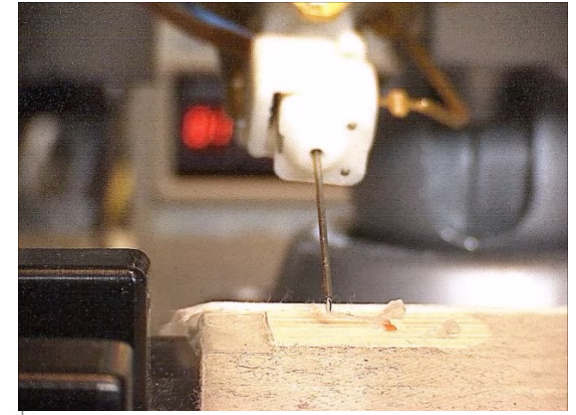
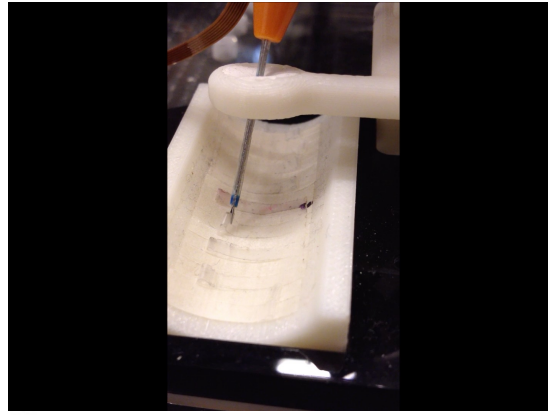
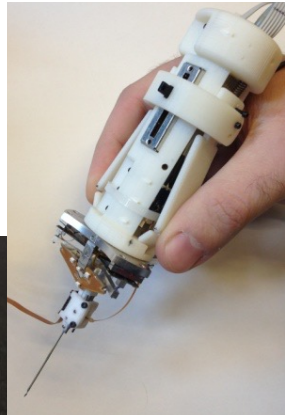




https://www.youtube.com/watch?v=JCpZ255_YVI&feature=emb_logo

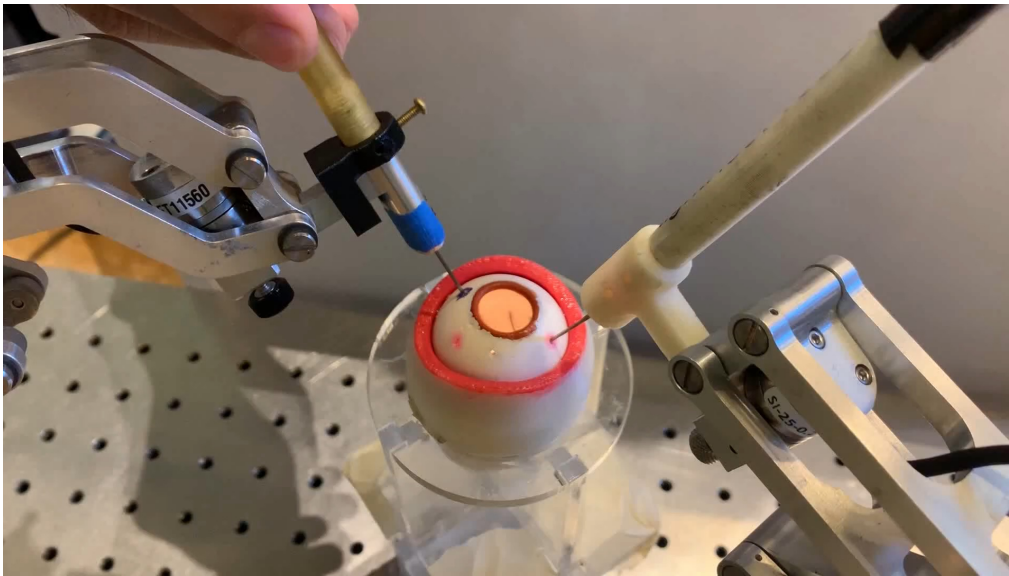


Example: Force-limited Retinal Membrane Peeling

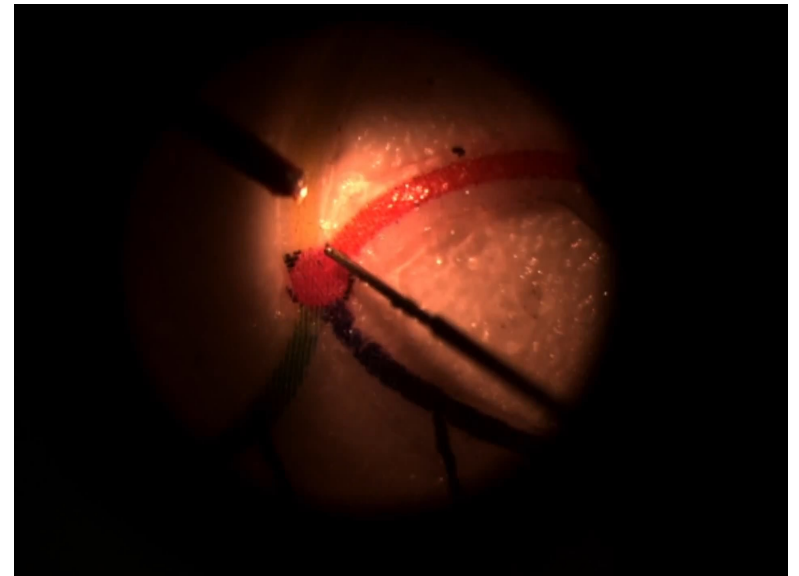


Automatic Light Pipe Actuating System in Bilateral Surgery

Automatic light pipe actuation



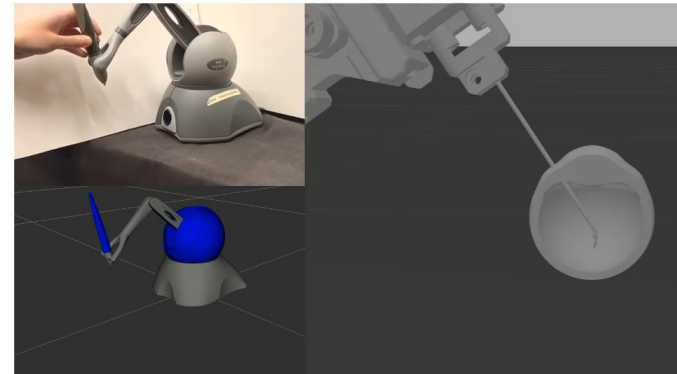
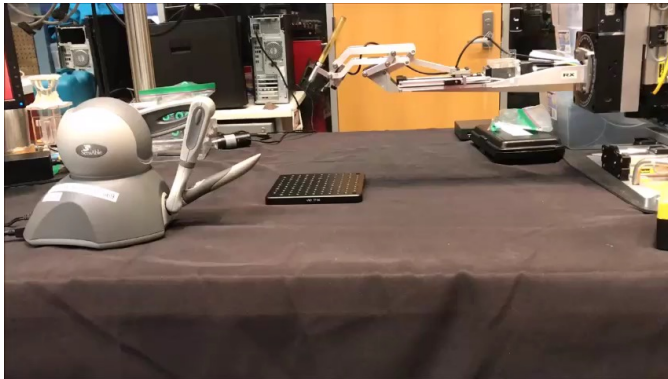
Microscope view



Changyan He, et al. *IEEE Trans. on Mechatronics*, 2020

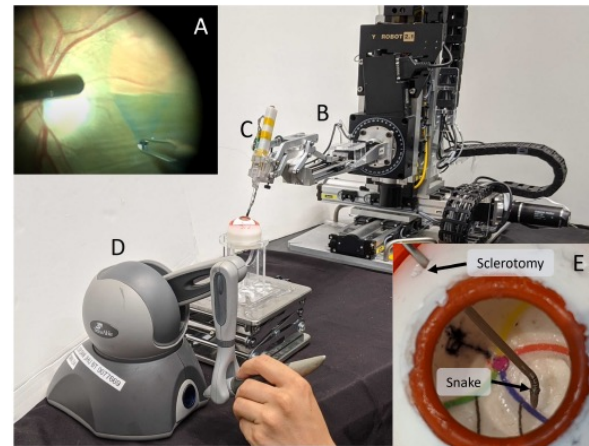
Teleoperation control of snake robot attached to eye robot

A. Ebrahimi, M. Jinno, I. Iordachita, *et al.*

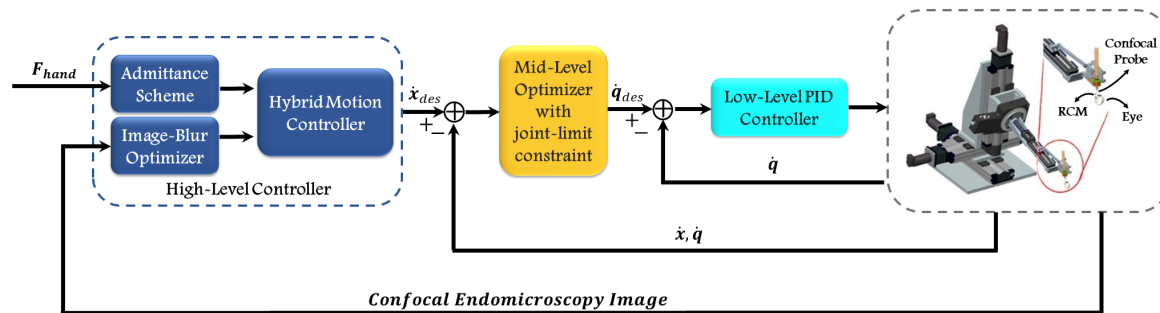


Envisioned high dexterity intraocular manipulator:

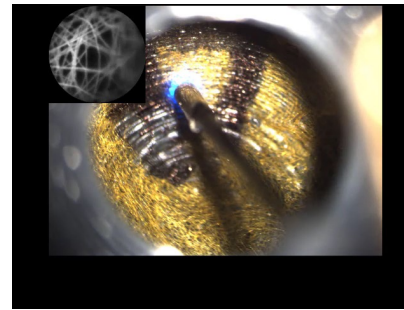
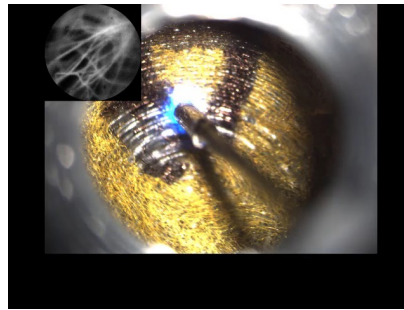
- (A) Epiretinal membrane peeling
- (B) Steady Hand Eye Robot
- (C) Integrated robotic intraocular snake robot
- (D) Phantom Omni
- (E) Distal snake-like tool-end inside eye phantom



Robot-assisted confocal endoscopic imaging for retinal surgery



Simple hand guiding with robot (5 DoF)



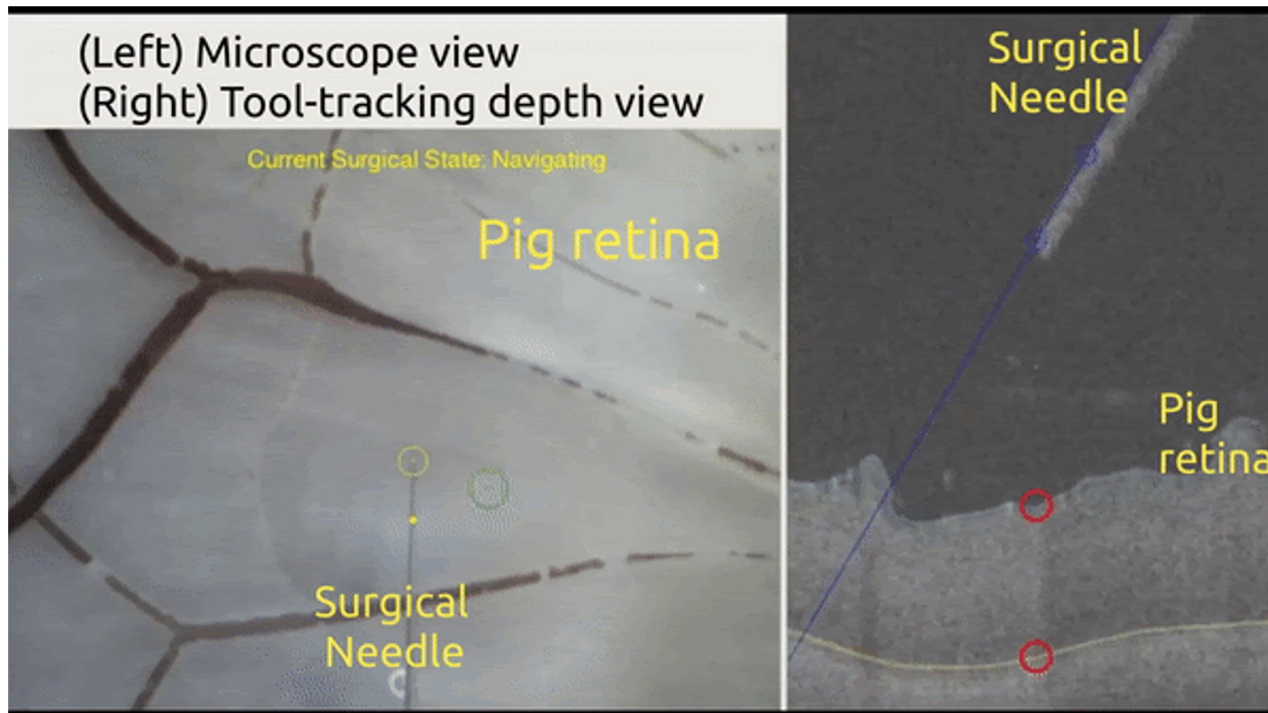
Hybrid control:

- Hand-guided lateral motion
- Image-based depth/focus control

Z. Li, M. Shahbazi, M. Patel, P. Chalasani, E. O'Sullivan, H. Zhang, K. Vyas, A. Deguet, P. Gehlbach, I. Iordachita, G. Z. Yang, R. H. Taylor, "A Comparison of Cooperative vs. Teleoperated Robot-Assisted Frameworks for Confocal Endomicroscopy Scanning of the Retina", *IEEE TMRB*, in submission.



Retinal Vein Cannulation with OCT + Video Guidance (10x speed)



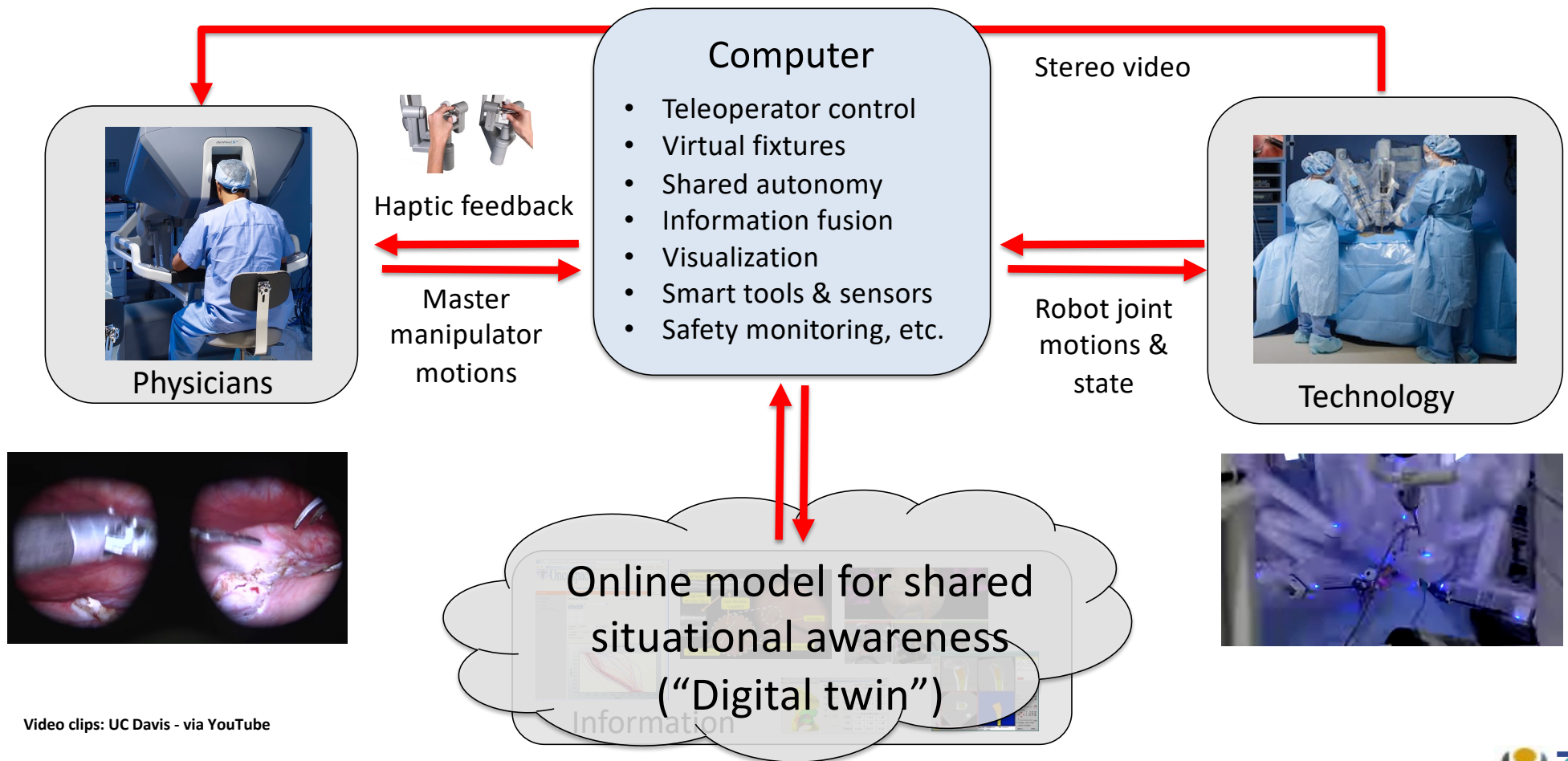
30 trials, 3 eyes (10 each)
Targeting accuracy: $26 \pm 12\mu\text{m}$
Total duration: $55 \pm 10.8\text{ s}$

A. Alamdar, D. E. Usevitch, J. Wu, R. H. Taylor, P. Gehlbach, and I. Iordachita,
"Steady-Hand Eye Robot 3.0: Optimization and Benchtop Evaluation for
Subretinal Injection", *IEEE Transactions on Medical Robotics and Bionics*, vol. 6-
1, pp. 135-145, 2024. 10.1109/TMRB.2023.3336975

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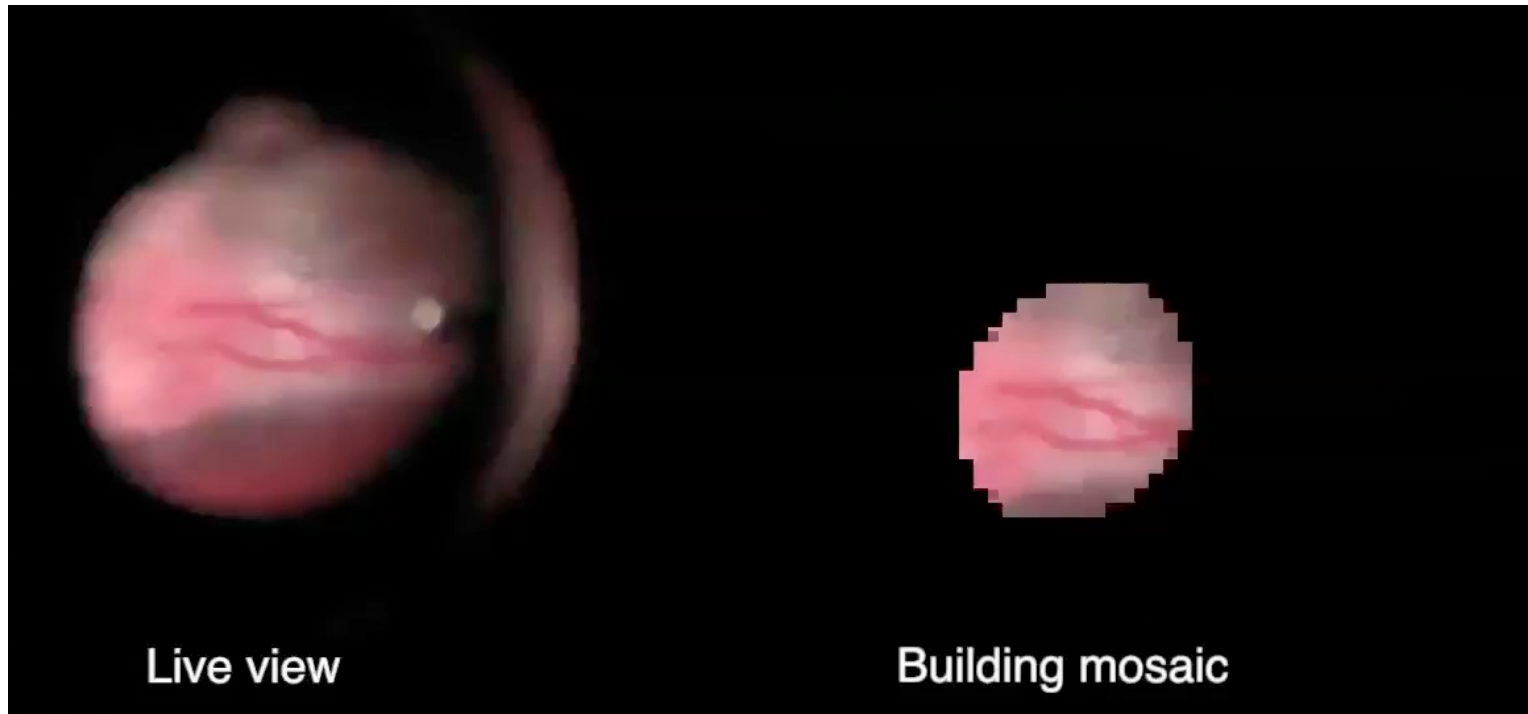
Laboratory for Computational Sensing and Robotics

Emerging paradigm (shared autonomy & assistant modes)

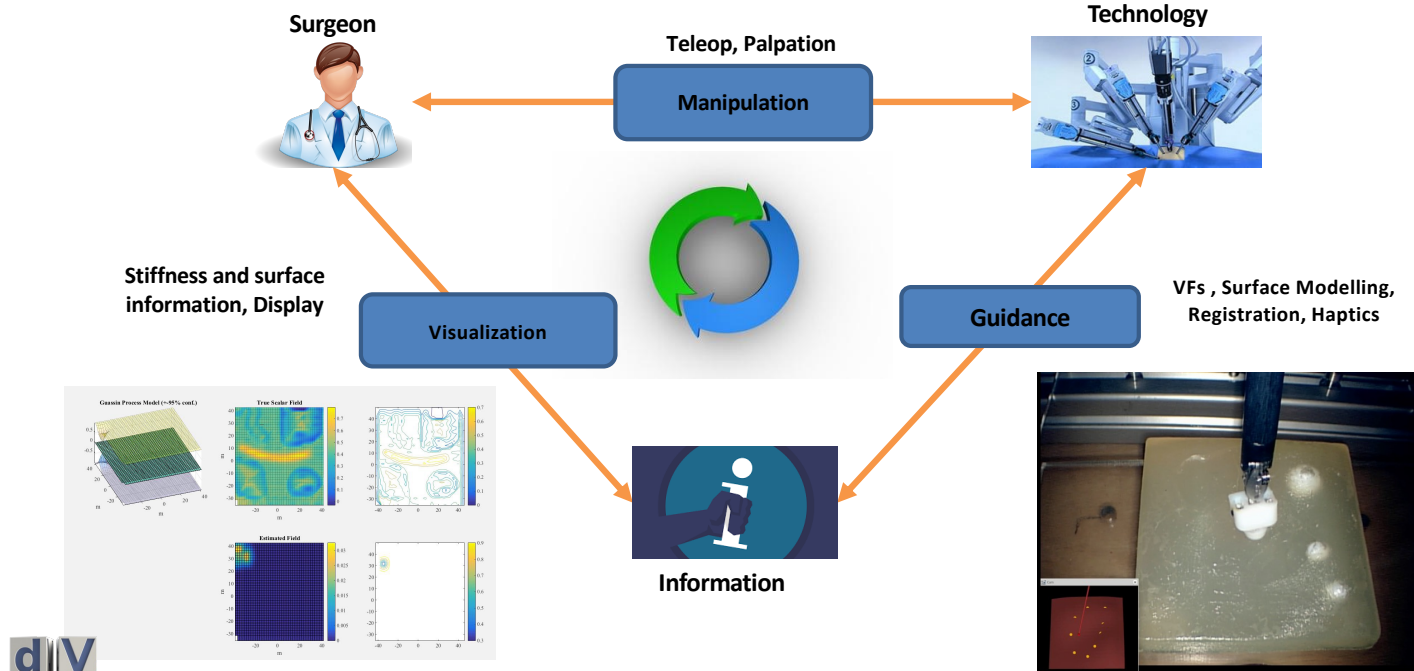
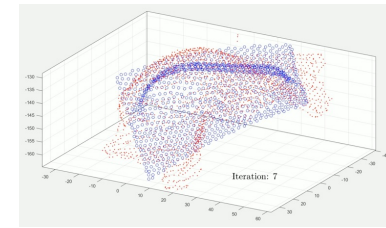
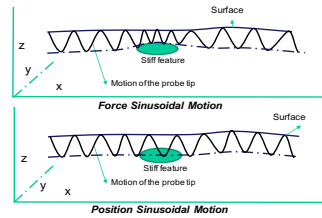


Video clips: UC Davis - via YouTube

Example: Retina Mosaicking, Annotation, and Registration

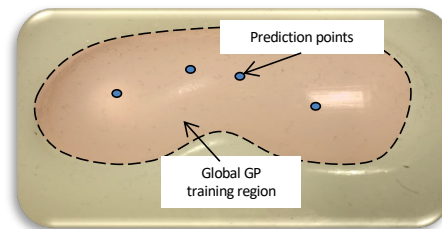


R. Richa, B. Vagvolgyi, R. Taylor, G. Hager, *MICCAI 2012*,



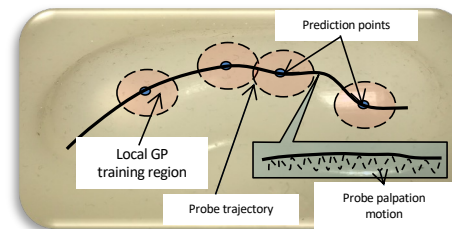
CONTINUOUS STIFFNESS AND GEOMETRY UPDATE

Preetham Chalasani, Russell Taylor



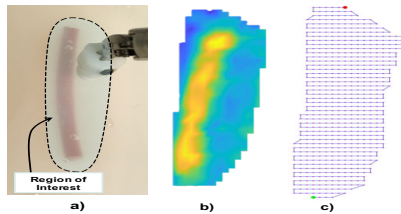
Offline Estimation

Chalasani et. al, ICRA 2016

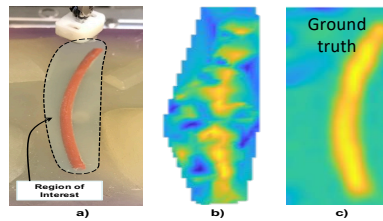


Online Estimation

Chalasani et. al, RAL 2018



Results from automated
Sinusoidal palpation

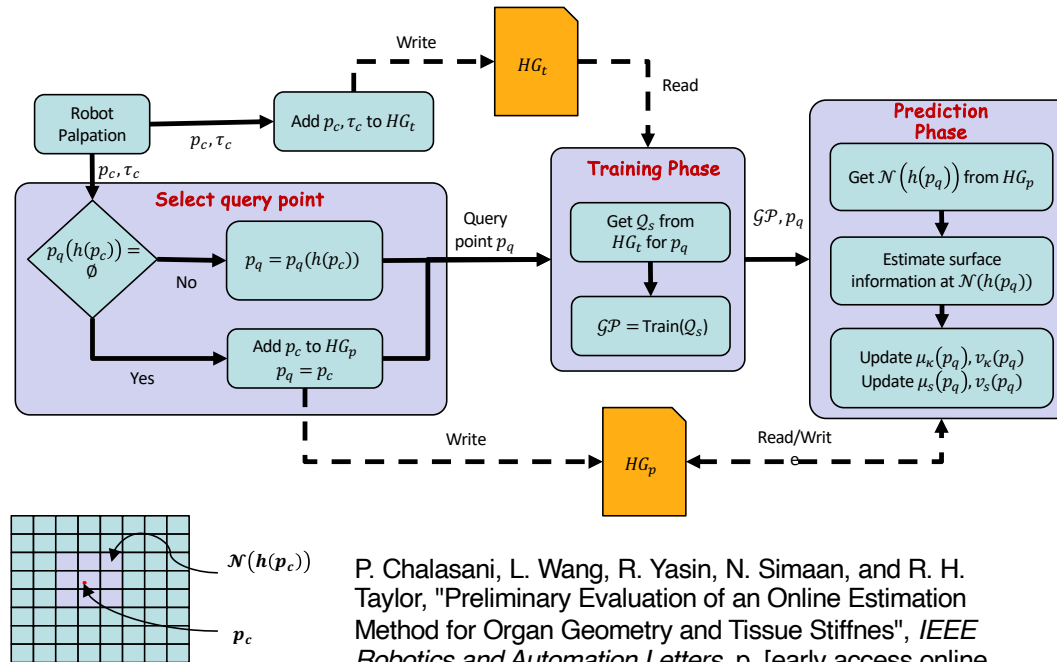


Teleoperated palpation w.
superimposed motion



Automated Sinusoidal Palpation

Online Geometry and Stiffness Update



P. Chalasani, L. Wang, R. Yasin, N. Simaan, and R. H. Taylor, "Preliminary Evaluation of an Online Estimation Method for Organ Geometry and Tissue Stiffnes", *IEEE Robotics and Automation Letters*, p. [early access online version], 2018. 10.1109/LRA.2018.2801481

Semi-Autonomous Palpation Example

Constrained Semi-Autonomous Telemanipulated Palpation with Assistive Virtual Fixtures

Preetham Chalasani, Long Wang, Rashid Yasin,
Peter Kazanzides, Nabil Simaan and Russell H. Taylor

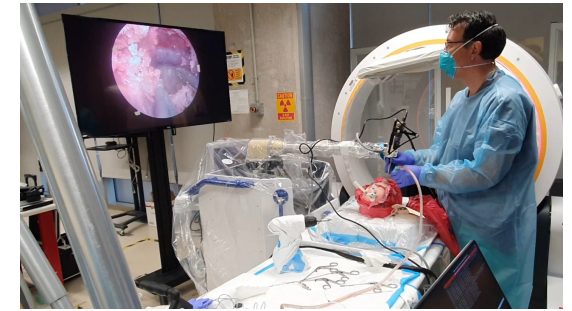


Example: Robot-Assisted ENT Surgery

Robotic neurotology surgery



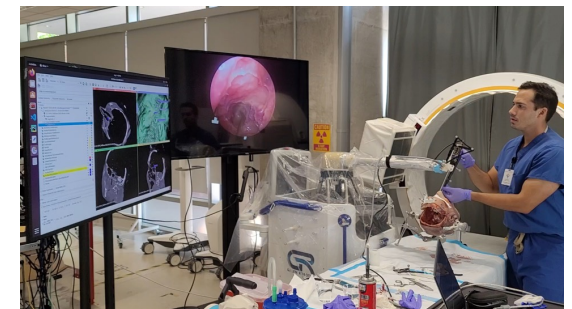
Robotic sinus surgery

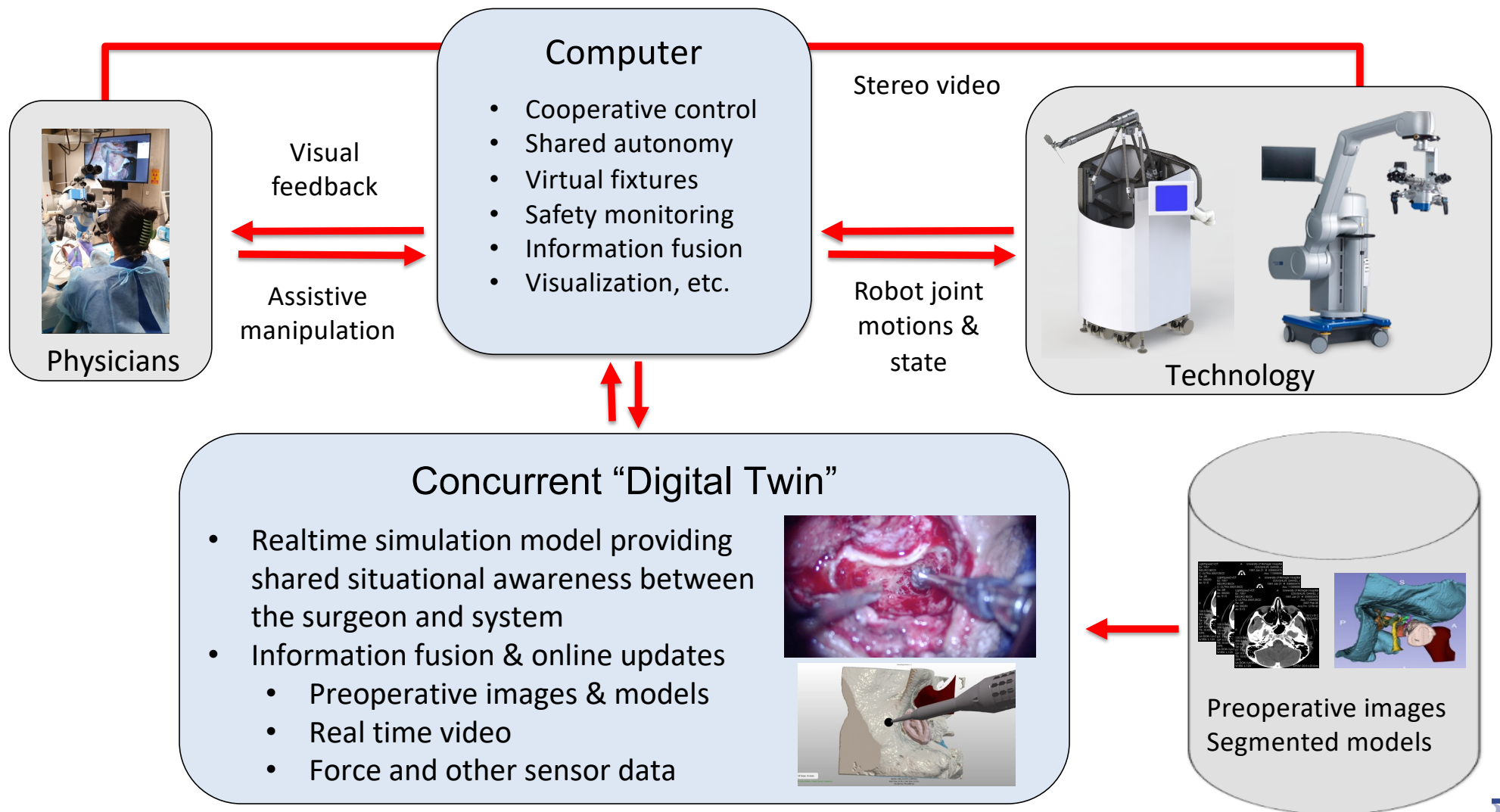


Robotic laryngeal surgery



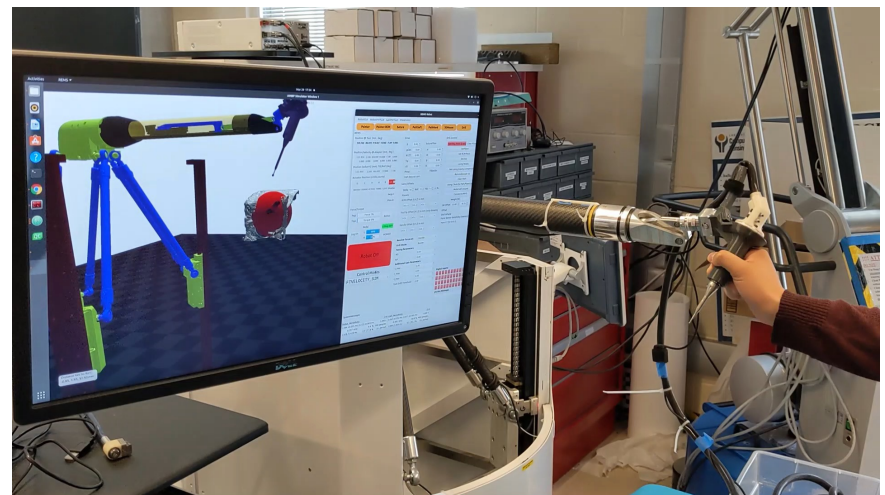
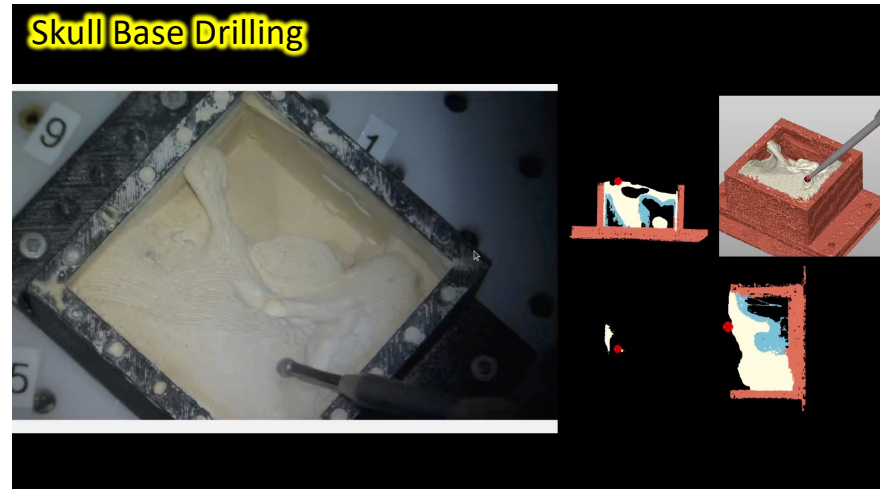
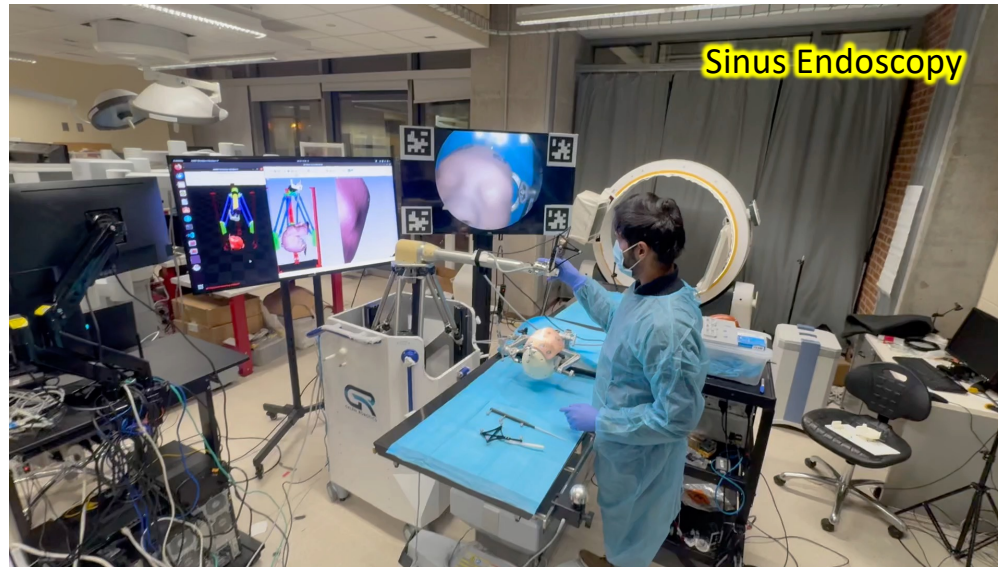
Robotic endoscopic endonasal skull base surgery



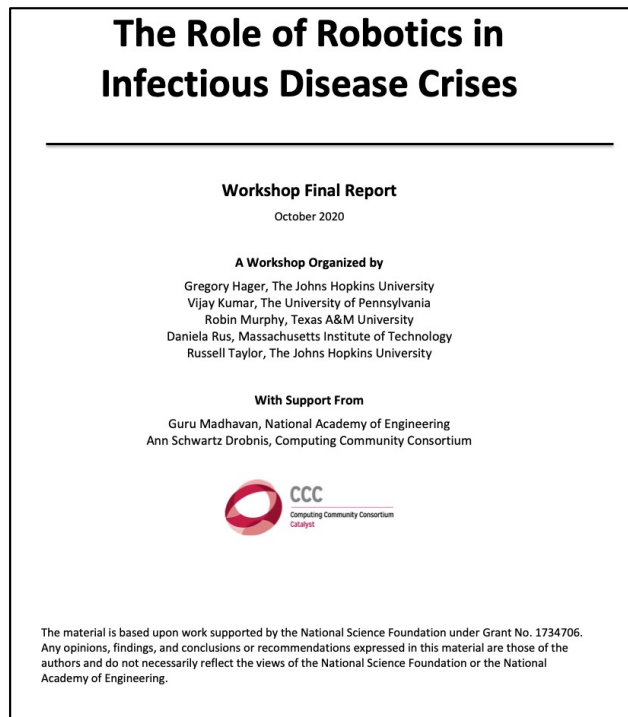


Digital Twin Examples

- Real-time simulation running concurrently with surgery
- Real-time sensor fusion and updates
- Provides shared situational awareness
- Information supports to surgeon & robot control



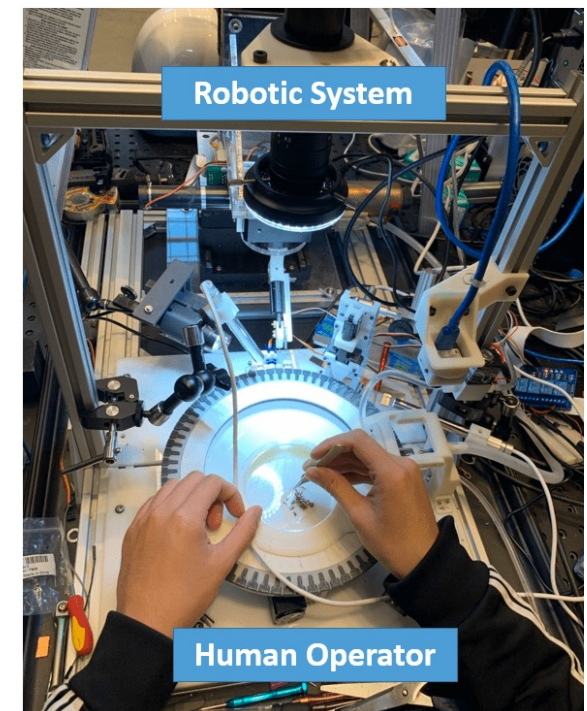
Beyond the Operating Room ...



NAE/CCC Workshop

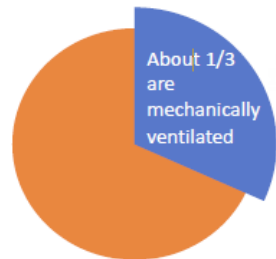


ICUs / Clinical Care



Vaccine Production

Assured Autonomous Mechanical Ventilation

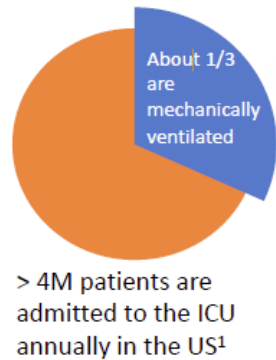


> 4M patients are admitted to the ICU annually in the US¹

- ICUs are **risk-laden** spaces
- Ventilator adjusted arbitrarily
 - **conservative** settings;
 - **longer** ventilation time
- Mechanical ventilation is tricky
 - **no gold standard** for patient care
 - **suboptimal** decision making

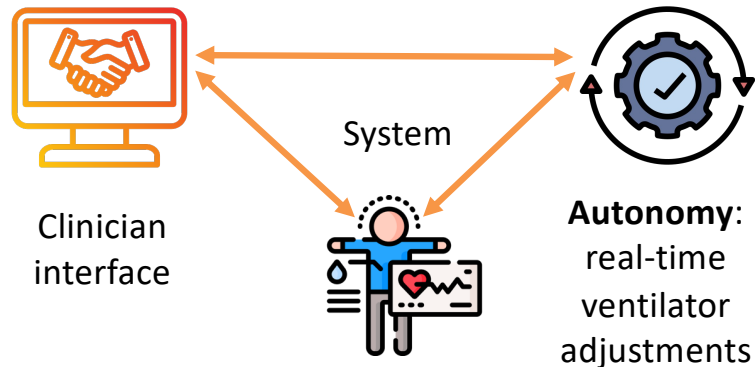


Assured Autonomous Mechanical Ventilation



- ICUs are **risk-laden** spaces
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Our proposed solution

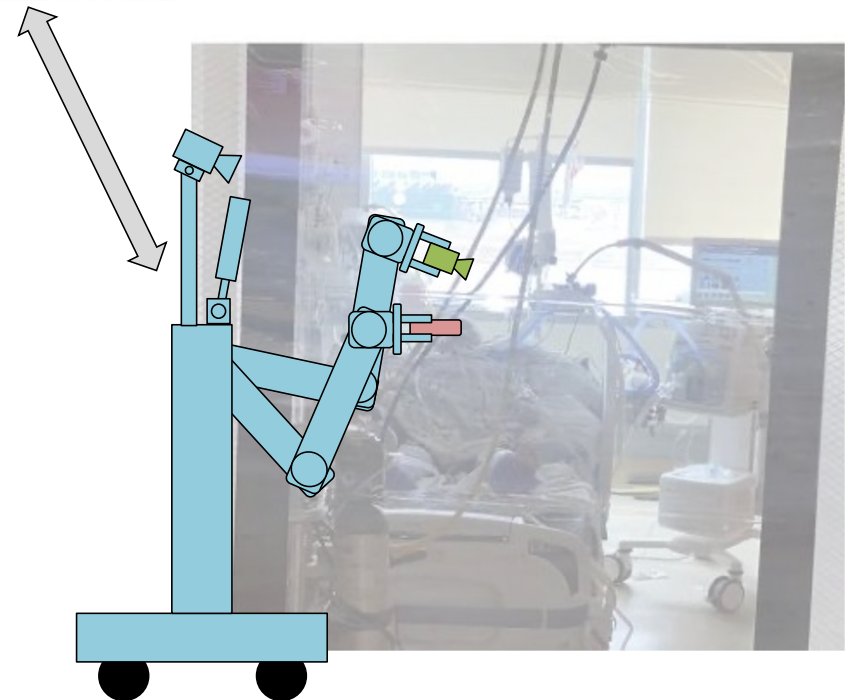


"Digital Twin": Online simulation of patient state based on multimodal sensing



Where this might go: Intelligent ICU assistant

- Multi-functional mobile robot with video links, robot arms, and a variety of tools and sensors
- Combination of teleoperation, shared, and autonomous control
- Sample tasks
 - Operate ventilators, infusion pumps, etc.
 - Feeding, delivering meals
 - Biohazard trash & sharps collection
 - Cleaning & disinfecting
 - Assisting care-givers during procedures
 - Surveillance & patient communication
 - Patient handling (may require different robot)



Key Challenges

- **Capability Challenges**

- Technology and devices
- Human-machine cooperation
- Information, situational awareness, trust

- **Deployability Challenges**

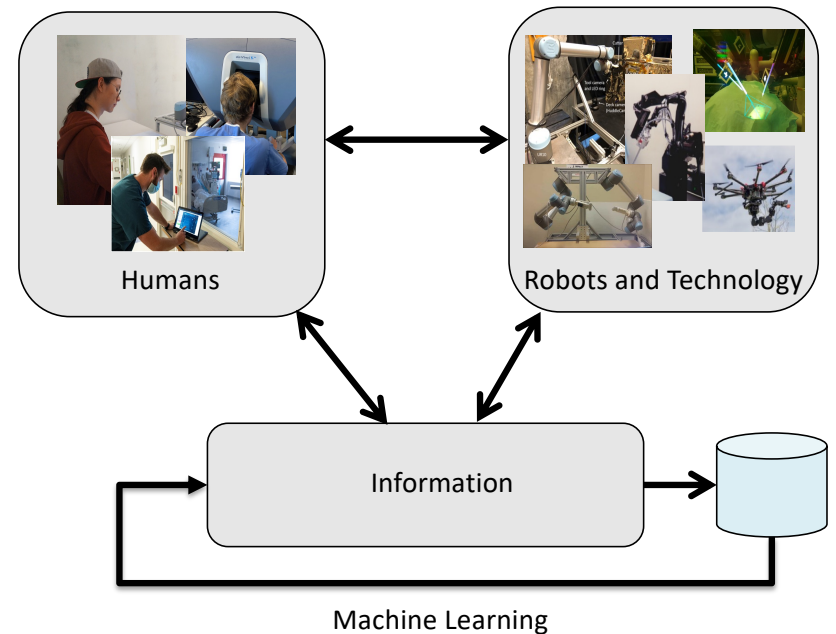
- Systems engineering, verification, validation
- Manufacturing, supply chain
- Training

- **Application-specific Challenges**

- Regulatory, sterility, disinfection, etc.

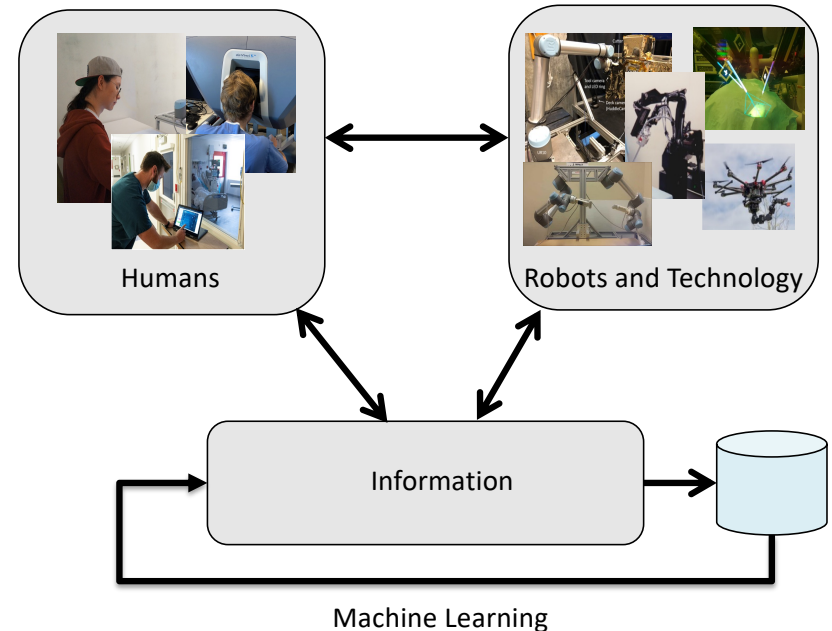
Technical Capability Issues

- Autonomy and intelligent systems
 - “Explainable” machine learning
 - Sensor/information fusion
 - Situational awareness
 - Safe actions in uncertain situations
- Human-machine communication
 - Speech, natural language, haptics, etc.
 - Augmented reality
 - “Trust” between robot and human on shared tasks
 - Programmability by non-expert users
- Hardware and physical capabilities
 - Sensors, actuators, mechanisms, etc.



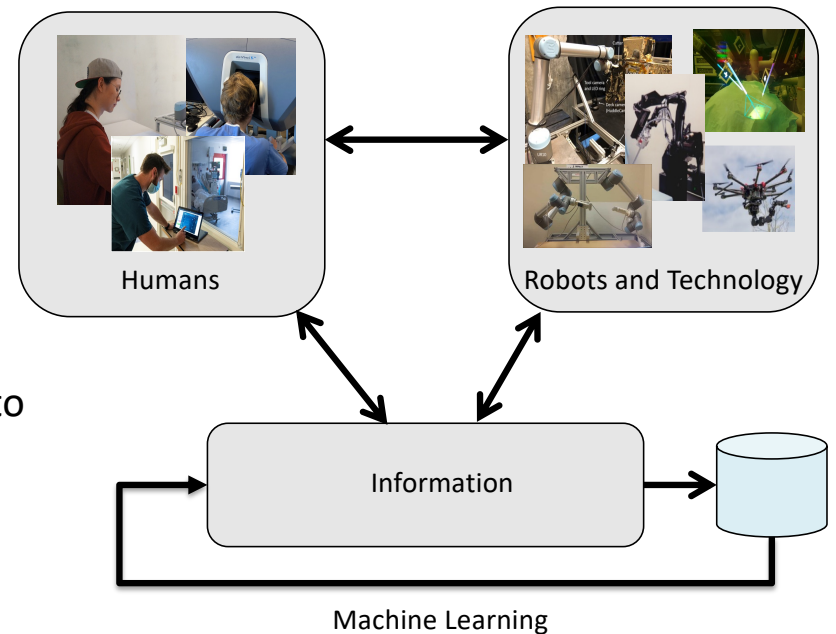
Systems Issues

- System engineering methods
 - System & interface design specification
 - Design verification and test
 - Fault detection, fault tolerance, and recovery
 - Risk assessment, projection, and mitigation
 - Need for “pre-certified” subsystems
 - **Added challenge:** Impact of AI and machine learning
- Testing and verification methods
 - Complex systems in unstructured environments
 - Need for formal and rigorous test methods
 - Non-expert users
 - “Normal” and unexpected “corner” cases
 - **Added challenge:** Adequacy of training data for machine learning components
- Agile and low-cost manufacturing



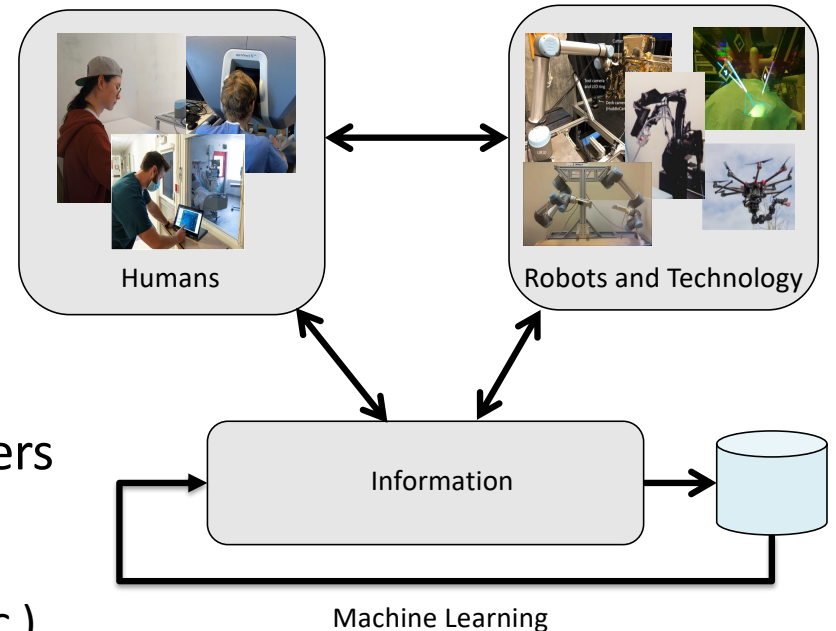
Mobilization and Infrastructure Issues

- Installed base
 - Need enough systems in the field so that repurposing is feasible.
 - These systems **need to be justified economically**.
 - But these systems need to be designed with interfaces permitting rapid adaptation to new needs.
 - Regulatory and reimbursement incentives are needed to promote this.
 - A data base of pre-certified designs and application modules could be invaluable.
- Training
 - Engineers, end users, other stake-holders
 - Pre-trained individuals
 - Better methods for just-in-time training

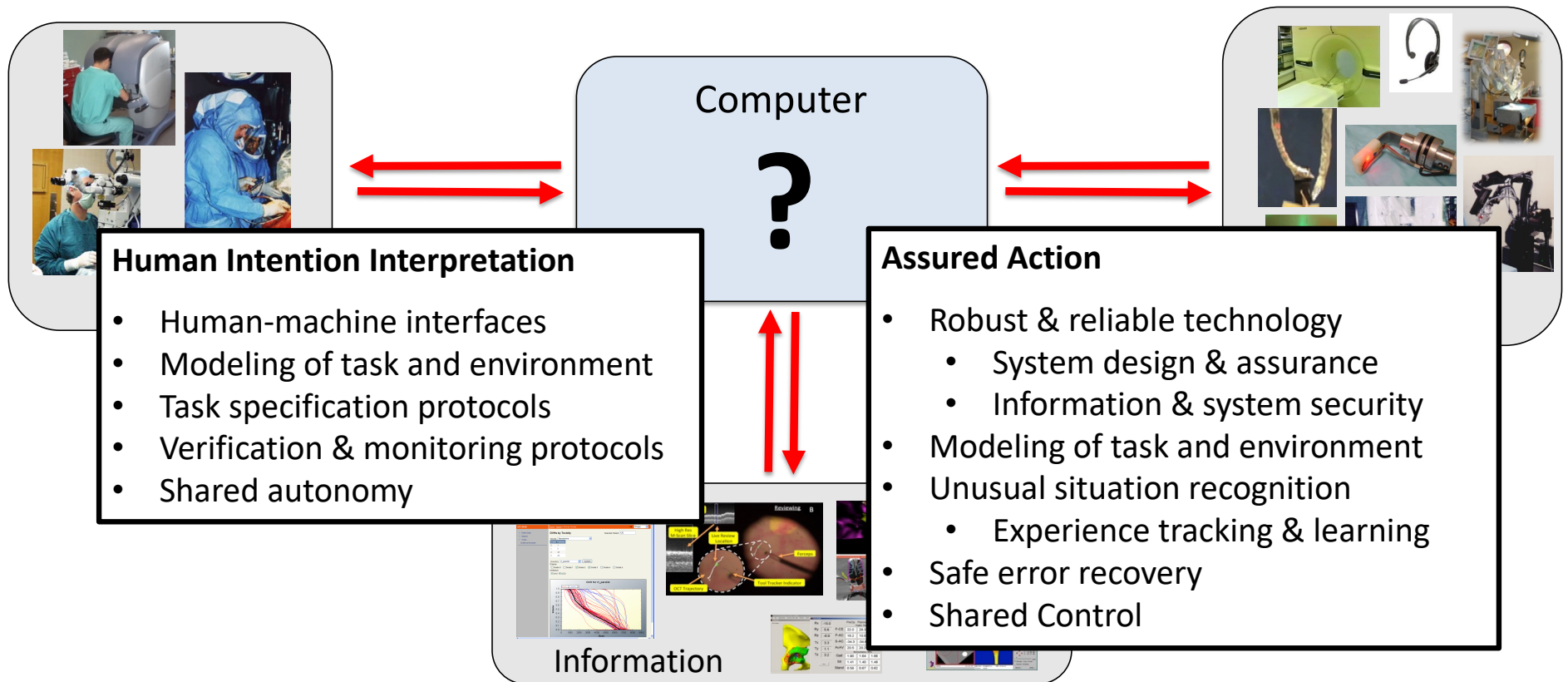


Application Domain-Specific Issues

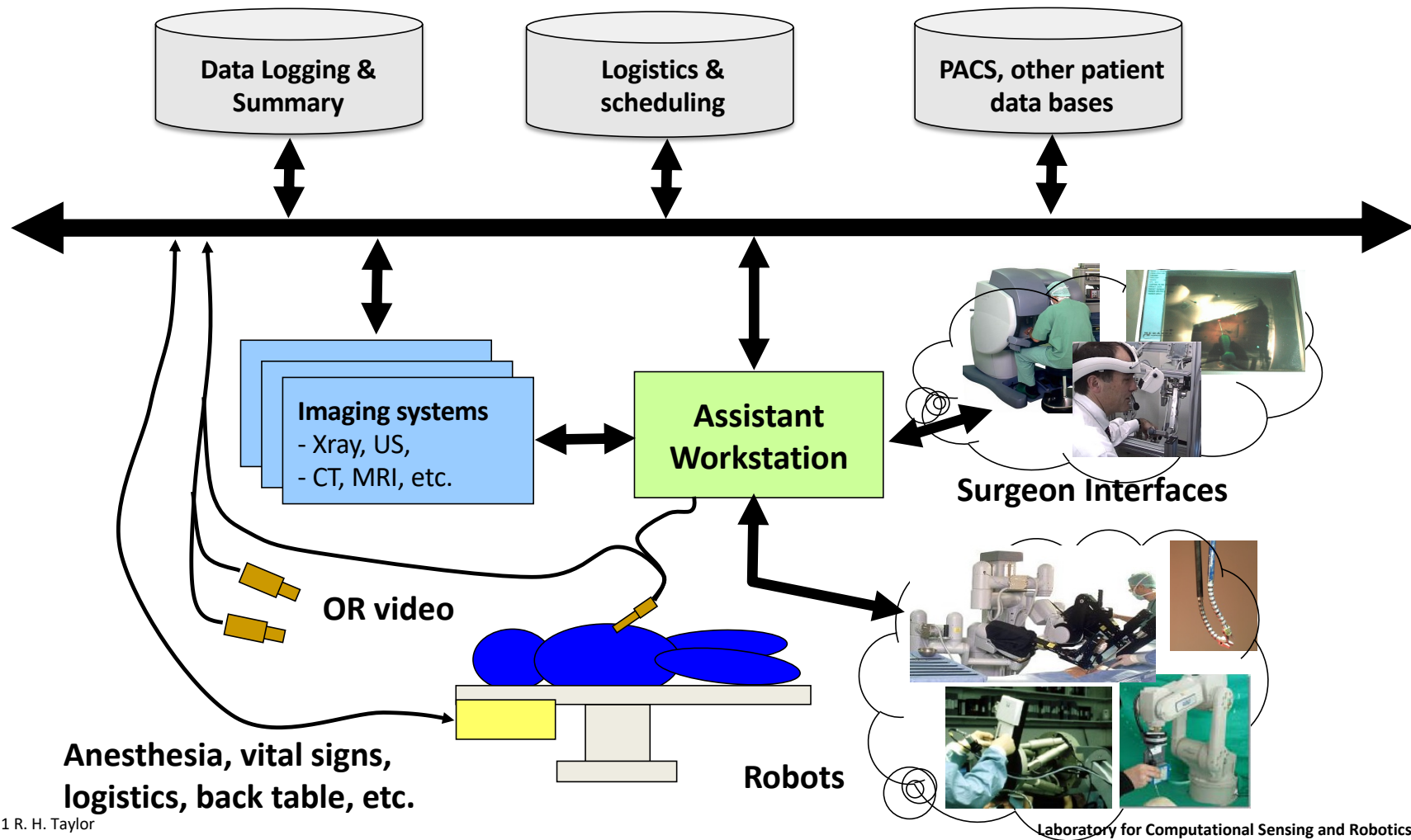
- Regulatory concerns
 - FDA
 - Institutional review boards
- Sterility & infection control
 - Current best practices
 - Need for advanced, self-disinfecting materials
 - Sensors and methods to detect infectious agents
- Specific needs and expectations of human users
 - Physicians
 - Nursing staff
 - Other support staff (hospital logistics, cleaning, etc.)
 - Patients
- Development of knowledge bases and communication channels



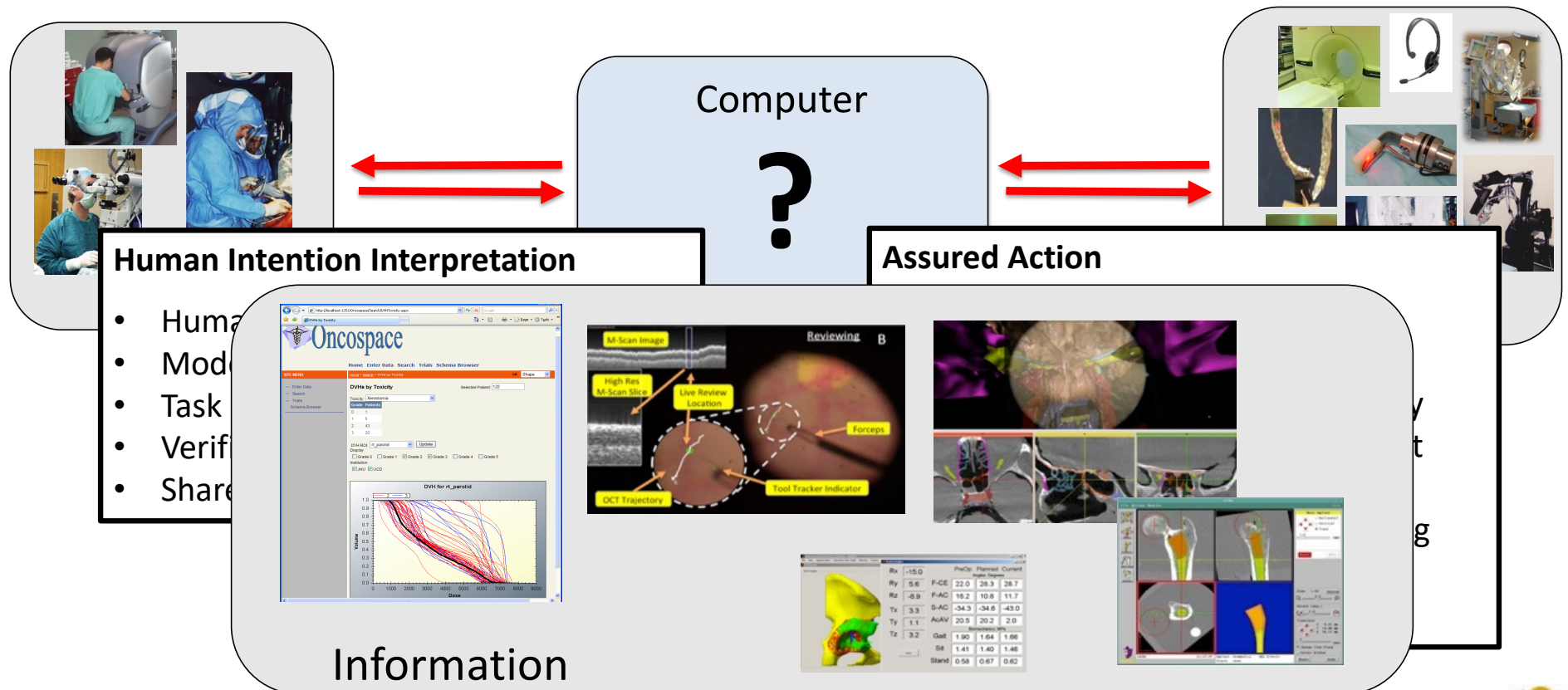
The key issue: mediating between human intention and action



Information-Intensive Interventional Suite



Information and machine “intelligence” are the crucial links



The real bottom line: patient care

- Provide new capabilities that **transcend human limitations** in surgery
- Increase **consistency and quality** of surgical treatments
- Promote **better outcomes** and more **cost-effective** processes in surgical practice

